

RESEARCH ARTICLE

CitNAC71-CitNAC76 coordinate cellulose and hemicellulose biosynthesis to regulate high-temperature-mediated granulation in citrus

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Short title: CitNAC71-CitNAC76 regulate citrus granulation

One-sentence summary:

High temperature flips a molecular switch that promotes cell wall polysaccharide accumulation, causing juice sac hardening and section drying in citrus fruit.

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Abstract:

Juice sac granulation, a globally pervasive physiological disorder in citrus (*Citrus* spp.), leads to substantial economic losses annually. The disorder is characterized by the marked accumulation of cell wall components, yet the molecular mechanisms coordinating this process remain poorly understood. Here, we found that high temperature acts as a strong inducer of citrus granulation, triggering a rapid and pronounced accumulation of cellulose and hemicellulose in juice sacs—an early hallmark of the disorder. Through integrated transcriptomic, biochemical, and genetic analyses, we identified *Cellulose synthase A 7/8* (*CitCesA7/8*) and *IRREGULAR XYLEM 9/9H/14H* (*CitIRX9/9H/14H*) as essential biosynthetic genes underlying this process. We further revealed that the NAM/ATAF/CUC (NAC) transcription factor CitNAC76 directly responds to high temperature and activates

CitCesA7/8 and *CitIRX9/9H/14H*, acting as a central regulatory hub of cellulose and hemicellulose accumulation during granulation process. Notably, we uncovered a molecular ‘brake’, *CitNAC71*, which suppresses *CitNAC76* expression and compromises its transactivation of *CitCesA7/8* and *CitIRX9/9H/14H*, forming a dual-layered feedback module that fine-tunes cellulose and hemicellulose biosynthesis under heat stress. Together, our study unveiled the *CitNAC71*-*CitNAC76* regulatory circuit that orchestrates stress-induced cell wall remodeling in citrus fruit, providing potential genetic targets for mitigating juice sac granulation and improving postharvest performance in a warming climate.

Keywords:

Citrus juice sac granulation;
Cellulose and hemicellulose;
Transcriptional regulation

1. Introduction

Citrus (*Citrus spp.*) is among the most widely cultivated and economically valuable fruit crops worldwide, with global production exceeding 160 million metric tons annually, serving as a vital source of nutrition and income for millions of farmers (FAO, 2023). However, juice sac granulation—a pervasive physiological disorder affecting major commercial species such as mandarin (*Citrus reticulata*), pomelo (*Citrus grandis*), orange (*Citrus sinensis*), and grapefruit (*Citrus paradisi*)—poses a severe threat to fruit quality and marketability. Granulation, characterized by juice sac hardening, reduced extractable juice, and nutritional depletion (Yao et al. 2018; Cao et al. 2020; Kang et al. 2022b), leads to significant economic losses, with reported incidence rates ranging from 12.2% to 62.5%

depending on cultivar and environmental conditions (Chen et al. 2005; Sharma et al. 2006; Li et al. 2022b). Alarming, granulated fruit remain visually indistinguishable from healthy fruit, complicating post-harvest sorting and exacerbating supply chain losses.

Granulation is triggered by multiple factors, including aberrant temperature conditions, plant hormonal imbalance and disruptions in reactive oxygen species (ROS) metabolism (Wu et al. 2020; Shi et al. 2020; Nie et al. 2020). For instance, low winter temperatures exacerbate granulation incidence in late-ripening navel oranges (*Citrus sinensis* (L.) Osbeck), while elevated temperatures during the flowering period intensify disorder severity (Ritenour et al. 2004; Wu et al. 2020). Similarly, Ponkan mandarin (*Citrus reticulata* Blanco) exhibited lower juice percentage, and higher weight loss and granulation rate when stored at high temperature (Lee et al. 2015). However, the underlying mechanism of high temperature-induced granulation still remains obscure. Given the increasing frequency of extreme temperature event under climate change, elucidating the regulatory basis of high temperature-induced granulation is essential for developing resilient citrus production strategies.

Granulation is accompanied by profound metabolic reprogramming in juice sacs, including decline in concentration of sugars, organic acids and amino acids, alongside the biosynthesis of cell wall components (Wang et al. 2014; Nie et al. 2020; Kang et al. 2022b; Li et al. 2022a). Substantial evidence indicates that granulation is tightly associated with aberrant cell wall remodeling, including pectin, cellulose, hemicellulose, and lignin (Hwang et al. 1990; Yao et al. 2018; Cao et al. 2020). Among these, lignification has been intensively studied as one of the crucial triggers of granulation (Chen et al. 2021; Jia et al. 2018; Li et al. 2022b). However, increasing evidence indicates that overaccumulation of cellulose and hemicellulose also plays a vital role in citrus granulation. For instance, granulated juice sacs of Marsh grapefruit exhibit twofold higher structural polysaccharides than healthy tissues (Hwang et al. 1990), and cold stress-induced granulation in navel oranges is associated with reduced cellulase activity and consequent cellulose accumulation (Wu et al. 2020). Histological and biochemical analyses have similarly revealed substantial accumulation of cellulose and hemicellulose in granulated juice sacs of *Citrus reticulata*

Shatangju and pummelo fruit (Cao et al. 2020; Shi et al. 2020). Notably, our previous study further showed that in some citrus species, such as Huyou (*Citrus changshanensis*), cellulose and hemicellulose accumulation can occur earlier than lignin deposition during the initial stage of granulation, suggesting that reinforcement of structural polysaccharides may represent an early event in granulation development (Kang et al. 2022c). Despite these observations, the molecular mechanisms and environmental signals underlying this early cell wall remodeling process remain largely unknown. In addition, because granulation develops internally within juice sacs, it is difficult to monitor dynamically in intact fruit. The lack of a stable and tractable experimental system has therefore been a major obstacle to dissecting the regulatory mechanisms underlying citrus granulation.

Cellulose and hemicellulose constitute 50-80% of plant cell wall polysaccharides, conferring mechanical integrity and resistance to environmental stress (Endler and Persson 2011; Kumar et al. 2016). The biosynthesis of cellulose and hemicellulose is driven by multi-subunit cellulose synthase complexes (CSC) and glycosyltransferases (GT43/47/8 families), and orchestrated by an evolutionarily conserved transcriptional regulatory cascade (Taylor-Teeple et al. 2015; Grantham et al. 2017; Anderson and Kieber 2020). In *Arabidopsis thaliana*, VASCULAR-RELATED NAC DOMAIN 6/7 (VND6/7) play crucial roles in vessel differentiation and secondary cell wall (SCW) formation (Kubo et al. 2005; Yamaguchi et al. 2008). Besides, NAC SECONDARY WALL THICKENING PROMOTING FACTOR 1/2/3 (NST1/2/3) are specifically expressed in fiber cells and function redundantly to promote SCW biosynthesis (Mitsuda and Ohme-Takagi 2008). Additional NAC transcriptional factors, including SECONDARY WALL-ASSOCIATED NAC DOMAIN 2/3 (SND2/3) and VND1/2/3/4/5, act downstream in the regulatory cascade and contribute to the activation of SCW biosynthetic genes (Hussey et al. 2011). These regulators cooperatively promote expression of secondary wall cellulose synthase genes (*CesA4/7/8*) and xylan biosynthetic *IRX* genes (e.g., *IRX9/9H* and *IRX14/14H*), thereby coordinating SCW deposition. Functional conservation of these regulatory modules has been observed across diverse plant species such as poplar (*Populus trichocarpa*), cotton (*Gossypium hirsutum*), rice (*Oryza sativa*), and maize (*Zea mays*) (Zhong et al. 2010a, 2011; Fang et al.

2020; Ye et al. 2018). While these NAC modules have been well characterized in model species, their roles in fruit tissue—and particularly during stress-induced cell wall remodeling—remains largely unexplored.

In this study, we established a robust *in vitro* juice sac granulation system based on high-temperature treatment, enabling precise capture of early molecular events associated with this highly dynamic process. Using integrated transcriptomic, genetic and biochemical analyses, we aimed to identify upstream regulatory factors responsible for temperature-induced granulation. Importantly, we uncovered a transcriptional regulatory module composed of CitNAC76 and its antagonist CitNAC71 that coordinates the expression of *CitCesA7/8* and *CitIRX9/9H/14H* under high-temperature stress. These findings provide mechanistic insight into the transcriptional control of cell wall remodeling in fleshy fruit under abiotic stress and identify genetic targets for mitigating juice sac granulation and improving fruit quality under changing climatic conditions.

2. Results

2.1 High temperature induces granulation and secondary cell wall accumulation in citrus juice sac

Juice sac granulation is a progressive physiological disorder that hard to be detected without destructive sampling. To enable controlled and continuous monitoring, we established an *in vitro* juice sac culture system that permits direct visualization of the granulation process. Using this system, we tested whether temperature acts as a regulatory factor in granulation process by exposing cultured juice sacs to 15°C, 20°C, 25°C, 30°C or 35°C. Among these treatments, the juice sacs subjected to 35°C exhibited the most severe granulation symptoms, characterized by prominent whitening and loss of transparency in the juice sacs (Fig. 1A; Supplementary Fig. S1A). Granulation level was further quantified by measuring juice sac firmness and whiteness index, two established indicators of granulation (Kang et al. 2022c). Both juice sac firmness and whiteness index were significantly increased under prolonged 35°C treatment compared with 15°C

treatment (Fig. 1B and C; Supplementary Fig. S1B and C), demonstrating that high temperature significantly promotes granulation in Huyou juice sacs. Importantly, intact Huyou fruit subjected to 35°C treatment displayed similar granulation phenotypes (Supplementary Fig. S2), and comparable responses were observed in Ponkan mandarin and sweet orange (*Citrus sinensis* (L.) Osbeck) (Supplementary Fig. S3), suggesting that high-temperature was a strong inducer of juice sac granulation among citrus species.

Overaccumulation of cell wall components is a hallmark of juice sac granulation. To examine temperature-dependent alterations in cell wall components, alcohol-insoluble solids (AIS), protopectin, water-soluble pectin, hemicellulose, cellulose and lignin were sequentially extracted and quantified at 0, 7, 14, 21, 35 and 49 d after treatments (Supplementary Fig. S4). Notably, the obvious increase in cellulose and hemicellulose contents in juice sacs under 35°C treatment occurred at 21 d post-treatment (Fig. 1D and E). We further tracked the spatiotemporal accumulation pattern of cellulose and hemicellulose, which observed that cellulose component was initially distributed in epicutis of the juice sacs and then accumulated within internal tissues during granulation (Fig. 1F). Consistently, fluorescence signals corresponding to xylan and xyloglucan epitopes significantly increased under 35°C treatment, while almost no xyloglucan (labeled by LM10) was detected in 15°C treatment (Fig. 1F). These results provided further morphological and biochemical evidence that high temperature induces cellulose and hemicellulose accumulation during granulation process.

2.2 *CitCesA7/8* and *CitIRX9/9H/14H* govern cellulose and hemicellulose biosynthesis in citrus juice sac

To unravel the molecular mechanisms underlying cellulose and hemicellulose biosynthesis during granulation process, we carried out transcriptome analysis and weighted gene co-expression network analysis (WGCNA) of juice sacs treated at 15°C and 35°C (Fig. 1G; Supplementary Fig. S5). Using cellulose and hemicellulose contents as traits, all the differentially expressed genes (DEGs) were classified into 15 modules (Supplementary Fig. S6). Notably, the genes in the blue module exhibited a highly positive

association with both cellulose and hemicellulose contents, with a coefficient of 0.61 ($P = 0.003$) and 0.55 ($P = 0.010$), respectively. Among these genes, *Ciclev10007321m.g* (*CitCesA7*) and *Ciclev10014155m.g* (*CitCesA8*) were identified as the most differentially expressed cellulose synthase-related genes. Additionally, three hemicellulose biosynthesis-related genes, *Ciclev10012036m.g* (*CitIRX9*), *Ciclev10008306m.g* (*CitIRX9H*), and *Ciclev10025401m.g* (*CitIRX14H*), showed strong upregulated trends (Fig. 1H and I). The study was repeated in a subsequent year, and the phenotypic traits, as well as the expression patterns of the candidate genes were similar (Supplementary Fig. S7). These data indicate that *CitCesA7/8* and *CitIRX9/9H/14H* could be the key genes involved in cellulose and hemicellulose biosynthesis during the high-temperature-mediated granulation.

Although the cellulose synthase (CESA) gene family has been extensively studied in model plants (Kumar et al. 2016), the functional roles of *CitCesA7* and *CitCesA8* remain poorly understood in citrus fruit. Accordingly, we carried out phylogenetic analysis to elucidate the evolutionary relationships of *CitCesA7* and *CitCesA8* homologs across various plant species (Supplementary Fig. S8A). We further conducted transient overexpression in citrus leaves (Supplementary Fig. S8B). The transcript levels of *CitCesA7* and *CitCesA8* were both increased compared to the empty vector (EV) control (Supplementary Fig. S8C and E), as well as the cellulose contents (Supplementary Fig. S8D and F). These observations confirmed that *CitCesA7* and *CitCesA8* serve as cellulose synthases in citrus. *IRREGULAR XYLEM* (*IRX*) genes play a crucial role in xylan synthesis, a major component of hemicellulose (Peña et al. 2007). It was found that *CitIRX9*, *CitIRX9H*, and *CitIRX14H* are closely related to *PtIRX9* (from *Populus trichocarpa*), *ZmIRX9H* (from *Zea mays*), and *PtIRX14H* (from *Populus trichocarpa*), suggesting that *CitIRX9/9H/14H* might be involved in xylan biosynthesis (Supplementary Fig. S9A). Additionally, the hemicellulose contents in citrus leaves increased after overexpressing *CitIRX9*, *CitIRX9H*, and *CitIRX14H* (Supplementary Fig. S9B-H). Taken together, this indicates that *CitIRX9*, *CitIRX9H*, and *CitIRX14H* are responsible for xylan synthesis in citrus.

2.3 CitNAC76 mediated the high-temperature induced activation of *CitCesA7/8* and *CitIRX9/9H/14H*

To identify the potential upstream regulators of *CitCesA7/8* and *CitIRX9/9H/14H*, we searched transcription factors (TFs) within the WGCNA modules whose expression levels showed significant positive or negative correlations with cellulose and hemicellulose content, and finally 17 TFs were selected for further analysis (Supplementary Fig. S10A). Subsequently, dual-luciferase assays were performed to evaluate the transcriptional activity of candidate TFs on the promoters of candidate genes. The results indicated that CitNAC76, a NAC domain-containing factor, could significantly activate the promoter activity of *CitCesA7/8* and *CitIRX9/9H/14H* (5-80-fold), while other TFs only exhibited a weak activation effect on the promoters of candidate genes (Fig. 2A and Supplementary Fig. S10B-D). Notably, expression profiling further revealed that *CitNAC76* transcript is almost undetectable in healthy juice sacs under normal temperature conditions, but is strongly and specifically induced by high temperature treatment (Supplementary Fig. S10A and S21B), highlighting its distinctive role as a heat-responsive activator of cell wall biosynthesis.

To elucidate the evolutionary relationships of CitNAC76 and other NAC TFs involved in secondary cell wall biosynthesis, we carried out an extensive phylogenetic analysis among various plant species (Supplementary Fig. S11). We found that all of the NACs positively involved in secondary cell wall biosynthesis could be divided into three families, including SECONDARY WALL-ASSOCIATED NAC DOMAIN (SND), VASCULAR-RELATED NAC-DOMAIN (VND), and NAC SECONDARY WALL THICKENING PROMOTING FACTOR (NST). Among all the NACs, CitNAC76 is closely related to AtVND2, belonging to the VND family that involved in xylem vessel formation in cotyledons (Tan et al. 2018). This suggests that CitNAC76 may also play a critical role in regulating secondary cell wall synthesis. To clarify the subcellular localization of CitNAC76, CitNAC76-eGFP was transiently expressed in *N. benthamiana* leaves and observed in the nucleus and cytoplasm, suggesting that it may act as a transcription factor (Fig. 2B). To validate the interactions between CitNAC76 and

the promoters of *CitCesA7/8* and *CitlRX9/9H/14H*, we performed a yeast one-hybrid assay (Y1H) (Supplementary Fig. S12). The results showed that Y1HGold cells grew normally on the SD/-Leu medium (supplemented with AbA) with the combination of *CitNAC76*-pGADT7, indicating *CitNAC76* could directly interact with the promoters of *CitCesA7/8* and *CitlRX9/9H/14H* (Fig. 2C).

To further explore the motifs that *CitNAC76* binds to on the promoters of *CitCesA7/8* and *CitlRX9/9H/14H*, we expressed and purified the *CitNAC76*-GST recombinant protein using *E. coli* BL21(DE3)pLysS for subsequent electrophoretic mobility shift assays (EMSA) (Supplementary Fig. S13A). Based on previous reports, three probes containing this NAC binding site (NACBS) were selected on each promoter region of target genes (Zhong et al. 2010b; Fu et al. 2021; Gong et al. 2021). We further incubated the *CitNAC76*-GST protein with all three 3' biotin-labeled probes, and detected one or two promoter-binding shifts with each promoter, indicating the formation of protein-DNA complex *in vitro* (Supplementary Fig. S13B-N). Moreover, the intensity of binding became weaker with the addition of higher concentrations of cold probes as competitors (Fig. 2D-I). As expected, a strong shifted band was found when the NACBS on the cold probes was mutated, confirming that *CitNAC76* specifically binds to the promoters of *CitCesA7/8* and *CitlRX9/9H/14H* via NACBS. This evidence indicate that *CitNAC76* can simultaneously activate *CitCesA7/8* and *CitlRX9/9H/14H* via directly binding to NACBS on their promoters.

2.4 *CitNAC76* is required for high-temperature-induced cellulose and hemicellulose biosynthesis in citrus juice sac

We conducted comprehensive experiments to elucidate the role of *CitNAC76* in cellulose and hemicellulose biosynthesis. Firstly, we overexpressed *CitNAC76* in citrus leaves, derived from three-month-old Huyou (*Citrus changshanensis*) seedlings (Fig. 3A-C). Compared to the EV group, the content of cellulose and hemicellulose was significantly higher in the *CitNAC76*-OE group (Fig. 3D and E). Additionally, the expression levels of downstream genes (*CitCesA7/8* and *CitlRX9/9H/14H*) all significantly increased after overexpression of *CitNAC76* (Fig. 3F). Besides, we generated *CitNAC76*-silencing citrus

seedlings using virus-induced gene silencing (VIGS) (Fig. 3G). As shown in Fig. 3H, the TRV lines of CitNAC76 were successfully generated after VIGS. Compared with the corresponding TRV control, the contents of cellulose and hemicellulose were significantly decreased in the *CitNAC76*-TRV lines (Fig. 3I and J). Moreover, silencing *CitNAC76* significantly decreased the spatial distribution of cellulose and hemicellulose, as well as the transcript abundance of *CitCesA7/8* and *CitIRX9/9H/14H* (Fig. 3K-M).

To reveal the biological functions of CitNAC76 on fruit segments, juice sacs separated from mature Huyou fruit were used for transient overexpression and RNA interference (RNAi) assays (Supplementary Fig. S14). We found that the juice sac firmness, cellulose and hemicellulose contents increased after overexpression of *CitNAC76*, indicating a higher granulation level (Supplementary Fig. S14A-E). Not surprisingly, the expression levels of *CitCesA7/8* and *CitIRX9/9H/14H* were upregulated after overexpression of *CitNAC76* (Supplementary Fig. S14F). In contrast, silencing of *CitNAC76* failed to induce cellulose and hemicellulose accumulation in juice sacs, even under high-temperature treatment (Supplementary Fig. S14G-L). This result indicates that CitNAC76 is required for the heat-induced activation of cellulose and hemicellulose biosynthesis during the juice sac granulation, consistent with its role as a key transcriptional regulator in this process.

2.5 CitNAC76 is a conserved activator of cellulose and hemicellulose metabolism in citrus, tomato, and *Arabidopsis*

To further validate the biological functions of CitNAC76, we generated stable overexpression lines in *Citrus sinensis* (L.) Osbeck cv. Jin Cheng (Fig. 4A-C; Supplementary Fig. S15A). As expected, cellulose and hemicellulose contents were significantly elevated in *CitNAC76*-OE plants compared with WT plants (Fig. 4D-E). Consistently, *in situ* cell wall labeling showed markedly stronger cellulose signals in *CitNAC76*-OE lines, as indicated by toluidine blue and calcofluor white staining (Fig. 4F). Hemicellulose deposition, visualized by LM10 and LM25 antibodies, was also substantially enhanced in *CitNAC76*-OE lines (Fig. 4F; Supplementary Fig. S15B). Consistently, the expression levels of cell wall biosynthetic

genes, including *CitCesA7/8* and *CitIRX9/9H/14H*, were significantly upregulated in *CitNAC76*-OE lines (Fig. 4G).

Citrus, as a perennial woody plant with long juvenile phase, presents technical challenges for stable transgenic fruit analysis. To overcome this limitation, we evaluated the function of *CitNAC76* in tomato by generating *CitNAC76*-OE transgenic lines (Fig. 4I and J; Supplementary Fig. S16A). We collected T₁ generation fruit at the red ripe stage, and found that the fruit pericarp of transgenic fruit was thicker than that collected from WT lines (Fig. 4H). Texture analysis demonstrated that *CitNAC76*-OE fruit maintained significantly higher firmness than WT fruit, both at the red ripe stage and after 21 d of storage (Fig. 4K; Supplementary Fig. S16B and C). Biochemical analysis showed that three independent *CitNAC76*-OE lines accumulated 18.02-48.6% higher cellulose and hemicellulose content than WT (Fig. 4L and M). Histochemical examination of pericarp sections revealed enhanced cellulose deposition in the outer and middle layers of *CitNAC76*-OE fruits while immunofluorescent assays confirmed increased hemicellulose accumulation (Fig. 4N; Supplementary Fig. S16D). RT-qPCR further revealed that *CitNAC76* overexpression upregulated the transcript levels of *SlCesA7/8* and *SlIRX9/9H/14* (Fig. 4O). Importantly, these alterations did not negatively affect other basic fruit quality traits (Supplementary Fig. S16E). We also assessed the biological function of *CitNAC76* in *Arabidopsis thaliana*, which observed that overexpression of *CitNAC76* significantly enhanced stem mechanical strength and erectness in *Arabidopsis* plants (Supplementary Fig. S17 A-D). Furthermore, the contents of cellulose and hemicellulose were significantly increased in *CitNAC76*-OE lines compared to WT lines (Supplementary Fig. S17 E-I). Collectively, these results indicate that *CitNAC76* plays a conserved role in regulating secondary cell wall formation across plant species.

2.6 *CitNAC71* forms a temperature-opposed regulatory module with *CitNAC76*

To uncover temperature-responsive regulators associated with *CitNAC76*, we performed a pull-down screening by incubating *CitNAC76*-GST with total proteins extracted from citrus callus. LC-MS/MS analysis combined with UniProt annotations identified 22

putative interacting proteins, including three previously uncharacterized NAC transcription factors (CitNAC7, CitNAC30 and CitNAC71) (Supplementary Table S1). Among them, CitNAC71 displayed markedly higher transcript abundance than CitNAC7 and CitNAC30 under both 15°C and 35°C conditions, and therefore was selected for further investigation (Supplementary Fig. S18). Similar to CitNAC76, CitNAC71 localized to both the nucleus and cytoplasm (Supplementary Fig. S19).

A direct physical interaction between CitNAC71 and CitNAC76 was consistently supported by multiple independent assays. In yeast two-hybrid assays, Y2HGold cells co-transformed with *CitNAC71* and *CitNAC76* turned blue in the presence of X- α -gal, suggesting CitNAC71 could physically interact with CitNAC76 (Fig. 5A). In the luciferase complementation imaging (LCI) assay, the chemiluminescent signal could be detected only when *CitNAC71* and *CitNAC76* (fused to the N or C-terminal of LUC) were co-expressed together in *N. benthamiana* leaves (Fig. 5B). Additionally, bimolecular fluorescence complementation (BiFC) assays showed that YFP signals can be detected in the combination of 2YC-*CitNAC71* and 2YN-*CitNAC76* (Fig. 5C). Furthermore, GST pull-down assay confirmed that CitNAC71 can interact with CitNAC76 *in vitro*, whereas the Co-IP assay reflected the interaction between CitNAC71 and CitNAC76 *in vivo* (Fig. 5D and E; Supplementary Fig. S20).

Strikingly, the transcript levels of *CitNAC71* and *CitNAC76* exhibit distinct temperature-dependent trends at 15°C and 35°C (Supplementary Fig. S21). This pattern suggested that CitNAC71 may function antagonistically to CitNAC76 in the regulation of cellulose and hemicellulose biosynthesis. To test this hypothesis, we conducted a dual-luciferase assay, using SK, *CitNAC71*-SK and *CitNAC76*-SK as effectors, with the promoters of *CitCesA7/8* and *CitIRX9/9H/14H* on the LUC vector as reporters. The result showed that *CitNAC71*-SK coupled with empty SK had no significant activating or inhibiting effect on the promoters of *CitCesA7/8* and *CitIRX9/9H/14H*, while *CitNAC76*-SK coupled with empty SK could significantly activate these promoters (Fig. 5F and G; Supplementary Fig. S22). However, when *CitNAC76*-SK co-infiltrated with *CitNAC71*-SK, this activating effect was significantly weakened, indicating that *CitNAC71*-SK inhibited the transcriptional activation

of *CitCesA7/8* and *CitIRX9/9H/14H* mediated by *CitNAC76*-SK (Fig. 5F and G; Supplementary Fig. S22). Together, these findings reveal that *CitNAC71* forms a temperature-dependent regulatory module with *CitNAC76*. Through competitive interaction, *CitNAC71* functions as a negative partner that attenuates *CitNAC76*-mediated activation of *CitCesA7/8* and *CitIRX9/9H/14H*, indirectly modulating cellulose and hemicellulose biosynthesis.

To elucidate the biological function of *CitNAC71* in citrus, we transiently overexpressed and silenced *CitNAC71* in citrus leaves (Supplementary Fig. S23A). Transient overexpression of *CitNAC71* markedly decreased the transcript abundance of *CitNAC76* and its downstream target genes *CitCesA7/8* and *CitIRX9/9H/14H* (Supplementary Fig. S23B and D), accompanied by a significant decrease in cellulose and hemicellulose contents (Supplementary Fig. S23C). Conversely, transient RNAi of *CitNAC71* substantially increased the expression level of *CitNAC76* and its downstream genes (Supplementary Fig. S23E and G), leading to elevated accumulation of cellulose and hemicellulose (Supplementary Fig. S23F). These results indicate that *CitNAC71* acts as a negative regulator of *CitNAC76*-mediated cellulose and hemicellulose biosynthesis.

2.7 *CitNAC71* functions as a molecular ‘brake’ to counteract *CitNAC76*-mediated cellulose and hemicellulose biosynthesis

In order to determine whether *CitNAC71* also represses *CitNAC76* at the transcriptional level, we cloned the promoter of *CitNAC76* into pGreenII 0800-LUC vector as a reporter, with the *CitNAC71*-SK as an effector. Dual-luciferase assays showed that the *CitNAC71* significantly suppressed the promoter activity of *CitNAC76* (Fig. 5H and I). Consistently, Y1H assays revealed that *CitNAC71*-pGADT7 enabled the Y1HGold cells harboring *CitNAC76pro*-pAbAi to grow in the presence of 100 ng·mL⁻¹ AbA, demonstrating a direct interaction between *CitNAC71* and the *CitNAC76* promoter (Fig. 5J). To further verify the DNA-binding specificity, we purified the *CitNAC71*-GST fusion protein and found that it could directly bind to Probe 1, 2 and 3 on the *CitNAC76* promoter (Supplementary Fig. S24A and B). Furthermore, the intensity of the shifted band gradually weakened with

1 increasing concentrations of unlabeled cold probes as competitors (Fig. 5K-M). However,
 2 the competition was abolished as use of mutated cold probes at the NAC binding sites
 3 (TACG, CGTA and CACG), confirming the specificity of the interactions between CitNAC71
 4 and the *CitNAC76* promoter (Fig. 5K-M). Taken together, the results from dual-luciferase
 5 assay, Y1H and EMSA indicated that CitNAC71 directly binds to the promoter of *CitNAC76*
 6 and represses its transcription, thereby attenuating CitNAC76-mediated activation of
 7 cellulose and hemicellulose biosynthesis.

8 To dissect the functional interactions between CitNAC71 and CitNAC76, we first
 9 performed transient overexpression assays in citrus juice sacs (Fig. 6A). After 6 d of
 10 overexpression, the transcript abundance of *CitNAC76* was significantly reduced in the
 11 *CitNAC71*-OE group, indicating a direct suppression of *CitNAC76* by *CitNAC71* (Fig. 6B-C).
 12 Consistent with the former results, *CitNAC76* overexpression strongly promoted juice sac
 13 firmness, enhanced cellulose and hemicellulose accumulation, and upregulated
 14 *CitCesA7/8* and *CitIRX9/9H/14H* (Fig. 6D-I). Conversely, overexpression of *CitNAC71*
 15 significantly reduced juice sac firmness, decreased cellulose and hemicellulose content,
 16 and suppressed expression of CitNAC76 target genes (Fig. 6D-I). Importantly, co-
 17 expression of *CitNAC76* and *CitNAC71* attenuated the CitNAC76-induced increase in
 18 cellulose and hemicellulose biosynthesis, indicating that CitNAC71 can reduce the
 19 transcriptional activation activity of CitNAC76 on its target genes (Fig. 6D-I).

20 We further examined the genetic interaction between CitNAC71 and CitNAC76 by
 21 generating single- and double-silencing citrus seedlings using virus-induced gene silencing
 22 (VIGS) (Fig. 7A). Efficient silencing of *CitNAC71* and *CitNAC76* was confirmed in the
 23 respective TRV lines (Fig. 7B and C). Compared with the TRV control, *CitNAC76* transcript
 24 level was significantly increased in the *CitNAC71*-TRV lines, whereas *CitNAC76* silencing
 25 did not affect *CitNAC71* expression (Fig. 7B and C). Consistent with these expression
 26 changes, silencing of *CitNAC76* significantly reduced cellulose and hemicellulose
 27 accumulation in citrus seedlings, as well as the expression levels of *CitCesA7/8* and
 28 *CitIRX9/9H/14H* (Fig. 7D-G). In contrast, *CitNAC71*-TRV lines showed the opposite trend,
 29 with elevated accumulation of both cellulose and hemicellulose and transcript levels of

target gene. Notably, simultaneous silencing of *CitNAC71* and *CitNAC76* abolished these opposite phenotypes, resulting in cellulose and hemicellulose levels comparable to the control line (Fig. 7D-G). Taken together, these results indicate that *CitNAC71* acts as a molecular ‘brake’ on *CitNAC76*, directly suppressing *CitNAC76* expression and thereby antagonizing *CitNAC76*-induced cellulose and hemicellulose biosynthesis in citrus.

3. Discussion

Citrus juice sac granulation is a highly dynamic process that remains largely undetectable without peeling the fruit, posing substantial challenges for investigating the molecular mechanisms underlying this disorder (Cao et al. 2020; Kang et al. 2022b). High temperature is a critical inducer of citrus granulation, especially under current global warming conditions (Ritenour et al. 2004; Lee et al. 2015). In this study, we established a stable granulation model involving high-temperature treatment of *in vitro* cultured juice sacs, facilitating precise observation and investigation of this highly dynamic process (Fig. 1A). Through integrated transcriptome, biochemical, and genetic analysis, we identified a temperature-responsive regulatory module centered on *CitNAC76* and *CitNAC71* that modulates cellulose and hemicellulose biosynthesis during citrus granulation.

Cell wall remodeling is a typical feature during citrus juice sac granulation. Previous studies have reported that overaccumulation of cellulose and hemicellulose contributes to the juice sac granulation in citrus fruit (Hwang et al. 1990; Cao et al. 2020; Nie et al. 2020). However, earlier work mainly focused on the degradation process of cellulose and hemicellulose involved in granulation (Wu et al. 2020; Chen et al. 2021), whereas the mechanisms underlying cellulose and hemicellulose biosynthesis remains largely unexplored. In plants, cellulose and xylan biosynthesis are mediated by cellulose synthase (CESA) complexes and GT43 family glycosyltransferase, respectively (Taylor et al 1999; Somerville 2006; Lee et al. 2010). Based on transcriptome analysis, we identified five genes (*CitCesA7/8* and *CitIRX9/9H/14H*), which are positively involved in cellulose and xylan biosynthesis (Fig. 1G-I). Additionally, functional verification of *CitCESA7* and *CitCESA8*

1 provides evidence for their roles in cellulose biosynthesis in citrus (Supplementary Fig. S8).
2 Furthermore, overexpression of *CitIRX9*, *CitIRX9H*, and *CitIRX14H* in citrus leaves resulted
3 in increased hemicellulose content, supporting their predicted β -1,4-xylosyltransferase
4 activity (Supplementary Fig. S9). These findings provide a mechanistic foundation for
5 understanding how cellulose and hemicellulose accumulate during granulation process
6 under heat stress.

7 NAC transcription factors constitute a major plant TF family, characterized by a
8 conserved NAC domain and a highly variable transcription regulatory domain (Han et al.
9 2023). NAC TFs play pivotal roles in various aspects of plant growth and development
10 (Mallory et al. 2004; Taylor-Teeple et al. 2015). Notably, accumulating evidence highlights
11 NAC TFs as central regulators of SCW biosynthesis (Zhong et al. 2010b; Zhong and Ye 2012;
12 Taylor-Teeple et al. 2015). Using integrated transcriptomic, genetic and biochemical
13 analyses, we identified the NAC transcription factor CitNAC76 as a direct activator of
14 *CitCesA7/8* and *CitIRX9/9H/14H* (Fig. 2). Phylogenetic analysis revealed that CitNAC76
15 clustered closely with AtVND1/2/3, a subgroup of NAC regulators that function as key
16 regulators of SCW biosynthesis (Kubo et al. 2005; Tan et al. 2018). Consistent with this
17 evolutionary relationship, our results indicate that CitNAC76 functions as a transcriptional
18 activator of secondary cell wall biosynthetic genes, thereby promoting cellulose and
19 hemicellulose accumulation during juice sac granulation. Similar functions of NAC
20 transcription factors have been reported in other species. For example, MaNAC1 in banana
21 (*Musa acuminata*) activates several cellulose synthase genes during fruit development (Yin
22 et al. 2023), and SmNST1/2 regulate SCW deposition in eggplant (*Solanum melongena*)
23 (Wang et al. 2024). Consistent with these conserved functions, stable overexpression of
24 *CitNAC76* enhances fruit firmness in tomato and stem erectness in *Arabidopsis* (Fig. 4H-O;
25 Supplementary Fig. S17), indicating that CitNAC76 retains the ability to activate SCW
26 biosynthesis in other plants. These findings suggest that CitNAC76 represents a promising
27 molecular target for modulating fruit firmness and extending postharvest shelf life, and
28 provide a transcriptional framework for manipulating cell wall components in fleshy fruit.

Previous works in *Arabidopsis* have revealed a complex transcriptional network governing SCW biosynthesis, in which transcriptional activators and repressors function together to maintain balanced control of cell wall formation (Zhao et al. 2008; Taylor-Teeple et al. 2015). Several NAC transcription factors have been reported to antagonize NST/SND/VND-type regulators to limit excessive SCW deposition (Taylor-Teeple et al. 2015; Liu et al. 2025). For example, XND1 represses SCW formation by suppressing NST1 expression and inhibiting NST1-mediated transactivation (Zhang et al. 2020), whereas VNI proteins negatively regulate VND activity through protein-protein interaction (Yamaguchi et al. 2010; Ailizati et al. 2021). Our results suggest that CitNAC71 represses CitNAC76 at both transcriptional level and protein interaction level. On the one hand, CitNAC71 physically interacts with CitNAC76 and forms transcriptionally inactive heterodimers, thereby attenuating CitNAC76-mediated activation of *CitCesA7/8* and *CitIRX9/9H/14H* (Fig. 5A-G). On the other hand, the VIGS and promoter assays indicate that CitNAC71 also represses *CitNAC76* transcription by directly binding to its promoter, providing a second layer of negative regulation (Fig. 5H-M; Fig. 7C). This dual repression suggests that CitNAC71 acts as a molecular ‘brake’ that prevents overaccumulation of cellulose and hemicellulose under normal temperature. Comparable NAC-mediated inhibitory mechanisms have been reported in other fruit systems. In loquat (*Eriobotrya japonica*), EjXND1 represses SCW biosynthesis through the EjHB1-EjPRX12 module and regulates low-temperature-induced lignification in fruit (Liang et al. 2025). In pear (*Pyrus spp.*), PbNAC71 negatively regulates *PbWAT1*, and its activity can be further modulated by E3 ligase-mediated protein degradation (Cong et al. 2024). These studies suggest that NAC repressors play conserved roles in limiting SCW deposition in fruit tissues. Notably, most previously characterized NAC repressors regulate SCW formation primarily through either transcriptional repression or protein interaction-based inhibition. By contrast, CitNAC71 integrates both regulatory modes to antagonize CitNAC76, providing an additional layer of control over SCW biosynthesis. Future studies will be required to determine whether CitNAC71 directly recruits transcriptional co-repressors, alters chromatin accessibility, or

interacts with additional regulatory proteins at the *CitNAC76* promoter, thereby repressing its transcription.

Temperature appears to play a central role in modulating the balance between *CitNAC71* and *CitNAC76*. Our data showed that *CitNAC71* expression in juice sacs exhibits a transient postharvest increase under normal temperature (15°C), whereas this induction is markedly suppressed under high temperature (35°C) ([Supplementary Fig. S21](#)). This result suggests that high temperature reduce *CitNAC71* expression by inhibits its normal postharvest transcriptional activation, thus weakening its repressive capacity on *CitNAC76*. Several mechanisms may contribute to the reduced inhibitory effect of *CitNAC71* under high temperature. One possible explanation is that heat-induced transcription factors compete with *CitNAC71* for binding at the *CitNAC76* promoter. Promoter analysis of *CitNAC76* identified one putative stress-response element (STRE) and three G-box motifs ([Supplementary Fig. S25](#)). G-box elements are recognized by transcription factors such as phytochrome-interacting factors (PIFs), which integrate environmental signals including light and temperature ([Leivar and Monte 2014](#); [Ohama et al. 2017](#)). Recruitment of heat-responsive transcription factors to these cis-elements could therefore enhance *CitNAC76* transcription under high temperature and reduce the *CitNAC71*-mediated repression. Alternatively, high temperature may influence the promoter-binding activity of *CitNAC71* itself. Heat stress can affect transcription factor activity through changes in chromatin association, protein stability, or DNA-binding affinity, thereby modifying promoter occupancy ([Jung et al. 2016](#); [Ohama et al. 2017](#)). Under high temperature, the ability of *CitNAC71* to bind the *CitNAC76* promoter may therefore be reduced. In addition, heat stress may affect the protein-protein interaction between *CitNAC71* and *CitNAC76*. Our data indicate that *CitNAC71* forms heterodimers with *CitNAC76* that suppress *CitNAC76* activity. Temperature-dependent changes in transcription factor localization or interaction dynamics have been reported for other NAC proteins, such as the rice NAC factor ONAC023 ([Chang et al. 2024](#)). Whether similar mechanisms operate in the *CitNAC71*-*CitNAC76* module remains to be determined. Taken together, these possibilities suggest that high temperature may weaken the inhibitory effect

of CitNAC71 and shift the regulatory balance of the CitNAC71-CitNAC76 module toward CitNAC76 activity. Future studies examining promoter occupancy, transcription factor binding dynamics, and protein stability under heat stress will help clarify how temperature modulates CitNAC71-CitNAC76 cascade during citrus juice sac granulation.

In conclusion, we propose a temperature-responsive regulatory module of CitNAC71-CitNAC76-*CitCesA/IRX* involved in cellulose and hemicellulose biosynthesis during granulation process (Fig. 8). At normal temperature, CitNAC71 imposes dual-layered repression on CitNAC76 to prevent excessive accumulation of cellulose and hemicellulose. Under high-temperature stress, reduced *CitNAC71* expression releases the 'brake' on CitNAC76, enabling CitNAC76-mediated transactivation of *CitCesA7/8* and *CitIRX9/9H/14H*, leading to the overaccumulation of cellulose and hemicellulose and promoting granulation. This study provides mechanistic insight into how environmental temperature signals are translated into transcriptional control of SCW biosynthesis in fruit tissues. Fine-tuning the CitNAC71-CitNAC76 module could mitigate excessive cell wall accumulation in juice sacs and alleviate granulation of citrus fruit, especially under the global warming conditions. Given the conserved role of CitNAC76 in promoting cell wall reinforcement, this regulatory module may also provide a useful strategy for improving fruit texture and postharvest performance in other fleshy fruit.

4. Materials and methods

4.1 Plant materials and growth conditions

Huyou (*Citrus changshanensis*) fruit used in this study was harvested on November 20, 2021 from a local orchard in Changshan county, Zhejiang province, China. For the *in vitro* culture of juice sacs, intact and uniform Huyou fruit were selected and sterilized with 70% ethanol for 10 min, followed by 1% (v/v) sodium hypochlorite solution (NaClO) for 30 min. After separating in the laminar flow hood, the juice sacs were cultured on the Murashige and Tucker (MT) Medium (Phytotech, USA, Product ID: M5525), supplemented with 80 g·L⁻¹ sucrose and 7.5 g·L⁻¹ agar (pH = 5.6), and were subjected to different

temperature treatment (15°C, 20°C, 25°C, 30°C and 35°C) using thermostatic incubators (Bluepard, China), with a constant humidity of 80%. The sampling points were set as 0 d, 7 d, 14 d, 21 d, 35 d and 49 d after treatment (Supplementary Fig. S26).

Arabidopsis (*Arabidopsis thaliana*), *Nicotiana benthamiana*, tomato (*Solanum lycopersicum*), and citrus (*Citrus sinensis*) plants were grown in a greenhouse under a 16 h of light/ 8 h darkness cycle (70% RH, 25°C). The light intensity was maintained at approximately 6,000 lux using LED bulbs (Gexinlai, China). All the samples were frozen in liquid nitrogen immediately and stored at -80°C for further analysis. At least three biological replicates were conducted for each group.

4.2 Physiological indexes determination

Fruit firmness and whiteness index were measured according to previously reported methods (Kang et al., 2022c). A total of 20 biological replicates were performed for each group in the following analysis. The firmness of citrus juice sacs and tomato fruit was determined using a texture analyzer (Texture Technologies, USA) equipped with a P2 probe, and the value of firmness was represented as maximum compression force (N). The whiteness index (WI) of juice sac was measured using a colorimeter (Hunterlab, USA) and calculated according to the formula (1):

$$WI = 100 - \frac{[(100 - L^*)^2 + a^{*2} + b^{*2}]}{2} \quad (1)$$

4.3 Measurement of cell wall components

For the measurement of cell wall components in fruit, the citrus juice sacs and tomato pericarp were lyophilized and ground into a fine powder for the quantification of cell wall components. Extraction, isolation and determination of pectin and lignin were performed as described by Cao et al. (2020) and Kang et al. (2022a). For the detection of cellulose and hemicellulose in plant material, including *Arabidopsis* and citrus leaves, samples were ground into fine powder in liquid nitrogen. The cellulose and hemicellulose were extracted and measured by Cellulose/Hemicellulose Content Assay Kit (Solarbio, China, BC4285,

BC4445) using visible spectrophotometry, with the detection wavelength of 620 nm and 540 nm, respectively. The contents of cellulose and hemicellulose were calculated using a standard curve and expressed as mg·g⁻¹ fresh weight or dry weight.

4.4 Cell wall *in situ* labeling and microscopy

Paraffin sections and labeling of plant tissue were conducted as reported by Huang et al. (2023) and Su et al. (2023). The embedded plant tissue was sectioned at 5 μm thickness using an ultramicrotome (Leica, Germany). Primary antibodies LM10 and LM25 (Plant Probes, UK) (1:50 in PBS) were used for the immuno-fluorescence labeling of xylan and xyloglucan, respectively, with Alexa Fluor 488 goat anti-rat IgG (H + L) (Invitrogen, USA, Product ID: A-11006) added as secondary antibody (1:200 in PBS). Cellulose was stained with toluidine blue (Solarbio, China, Product ID: G3668) and calcofluor white (Coolaber, China, Product ID: SL7204) according to the manufacturers' instructions. After staining, the sections were photographed using Olympus BX61 (Olympus, Japan) at FITC (excitation wavelength, 488 nm), DAPI (excitation wavelength, 405 nm), and bright field, respectively. Autofluorescence observation for paraffin sections was shown in (Supplementary Fig. S27). The fluorescence intensity was calculated using ImageJ software (National Institutes of Health, USA) and presented as Mean gray value, according to the following formula (2).

$$\text{Mean gray value (A.U.)} = \frac{\text{Integrated Density}}{\text{Measured area}} \quad (2)$$

4.5 RNA extraction and sequencing

Total RNA was extracted using VeZol-Pure Total RNA isolation Kit (Vazyme, China, Product ID: RC202-01) according to the manufacturer's protocol. The library construction and RNA sequencing were performed by BGI (Wuhan, China). The constructed libraries were sequenced using HiSeq 4000 platform (Illumina, USA) to obtain clean raw reads. After filtering with SOAPnuke (Version 1.5.2), all clean reads were mapped to the reference genome of *Citrus clementina* (<https://www.citrusgenomedb.org/>). The gene abundances were calculated and normalized to the values of fragments per kilobase of transcript per

million mapped reads (FPKM). Differential expression analysis was performed as previously described (Kang et al. 2022c); the differentially expressed genes (DEGs) between two groups were restricted with a false discovery rate (FDR) < 0.05 and $|\log_2(\text{fold change})| > 1$. DEGs expression and WGCNA were calculated by clusterProfiler R software (Langfelder and Horvath 2008).

4.6 Reverse-transcription quantitative PCR (RT-qPCR)

The first-strand cDNA was synthesized using the HiScript III RT SuperMix for qPCR (+gDNA wiper) (Vazyme, China, Product ID: R323-01). RT-qPCR was performed with CFX96 Real-Time System (BIO-RAD, USA) using QK Platinum SYBR Green Master Mix (Thermo Fisher Scientific, USA, Product ID: A57156). Genes of *CitACTIN*, *AtEF1 α* , and *SlACTIN* were used as internal reference for citrus, *Arabidopsis*, and tomato, respectively. The relative gene expression levels were calculated by the $2^{-\Delta\Delta C_t}$ method. Primers used for RT-qPCR are listed in Supplementary Table S2, which were validated by melting curves and PCR products sequencing.

4.7 Gene cloning and phylogenetic analysis

Full-length coding sequences (CDS) of genes (*CitNAC76*, *CitCesA7*, *CitCesA8*, *CitIRX9*, *CitIRX9H*, and *CitIRX14H*) were amplified from juice sacs of Huyou (*Citrus changshanensis*) containing a mixture of different developmental stages. Homologous protein sequences were retrieved from TAIR database (<https://www.arabidopsis.org/>) and UniProt database (<https://www.uniprot.org/>). Multiple sequence alignment was performed using MEGA software (Version 11.0.13, USA) with default parameters. The phylogenetic trees were constructed via the maximum-likelihood method in MEGA software and visualized using iTOL (<https://itol.embl.de/>). The numbers of GenBank accession and UniProt entry used in phylogenetic trees can be found in Supplementary Table S3, S4 and S5. The multiple sequence alignments used to construct the phylogenetic tree are provided in Supplementary Files S1/S3/S5, and the tree files are provided in Supplementary Files S2/S4/S6.

4.8 Subcellular localization assay

The full-length CDS of genes (*CitNAC76* and *CitNAC71*) were inserted into the pCambia1300-eGFP vector, and the constructs were introduced to *Agrobacterium tumefaciens* (strain GV3101). Primers used are listed in [Supplementary Table S6](#). The *A. tumefaciens* with eGFP constructs were cultured on Luria-Bertani (LB) plates and resuspended in the infiltration buffer (10 mM MgCl₂, 10 mM MES, 150 μM acetosyringone; pH = 5.6) to OD₆₀₀ = 1.0, then injected into *N. benthamiana* leaves (with stable RFP nuclear localization marker) ([Liu et al. 2020](#)). After 2 d of injection, the fluorescence of GFP and RFP was observed and captured using a Zeiss LSM 710 NLO confocal microscopy (Zeiss, Germany).

4.9 Dual-luciferase assay

The transactivation activities of candidate transcription factors were tested by dual-luciferase assay as previously described ([Zhao et al. 2021](#)). Briefly, the coding sequences of transcription factors were inserted into the pGreenII 62-SK vector and the promoter fragments of *CitCesAs* and *CitIRXs* were inserted into pGreenII 0800-LUC vector. Primers used are listed in [Supplementary Table S7](#). The constructs were then introduced into the *A. tumefaciens* (strain GV3101) containing pSoup-P19 plasmid. The infiltration buffer containing *A. tumefaciens* was prepared as described in subcellular localization (OD₆₀₀ = 0.75) and injected into *N. benthamiana* leaves, with *A. tumefaciens* cultures containing SK and LUC constructs mixed in a ratio of 10:1 (v/v). After 3 d of injection, the signals of Firefly Luciferase (LUC) and Renilla Luciferase (REN) were detected with Dual-Luciferase Reporter Assay System (Promega, USA, Product ID: E1980).

4.10 Prokaryotic expression and electrophoretic mobility shift assays (EMSA)

For the expression of recombinant proteins, the CDS of *CitNAC71* and *CitNAC76* (without a stop codon) was cloned into the vector pGEX-4T-1, and the fusion constructs were introduced into *E. coli* strain BL21(DE3)pLysS. The empty GST and *CitNAC76*-GST

recombinant proteins were induced and collected as previously reported (Fu et al. 2021). Then the expressed proteins were purified using GST-tag Protein Purification Kit (Beyotime, China, Product ID: P2262), and confirmed by SDS-PAGE analysis. Primers are listed in [Supplementary Table S8](#).

The EMSA analysis was conducted using LightShift Chemiluminescent EMSA kit (Thermo Fisher Scientific, USA, Product ID: 20148) following the manufacturer's instructions. Oligonucleotide probes containing the NACBSs were labelled with 3' biotin by GeneBio Technology (GeneBio, China). DNA-protein complexes were separated by 5% (w/v) native PAGE on ice (160 V, 30 min). After being transferred to a positively charged nylon membrane and cross-linked by UV light, the DNA-protein complexes were detected by chemiluminescent nucleic acid detection module and photographed by ChemiDoc XRS+ System (BIO-RAD, USA). All the EMSA experiments were repeated at least three times.

4.11 Yeast one/two-hybrid assays

For Yeast one-hybrid assay (Y1H), the promoters of target genes (*CitNAC76*, *CitCesA7*, *CitCesA8*, *CitIRX9*, *CitIRX9H*, and *CitIRX14H*) were ligated into the pAbAi vector as bait constructs, while the full length of *CitNAC71* and *CitNAC76* were inserted into the pGADT7 vector as prey constructs. Then, the pGADT7 vectors were transformed into Y1HGold competent cells harboring the pAbAi-bait constructs. Interactions between prey and bait constructs were tested on synthetic defined medium lacking Leu (SD/-Leu), supplemented with aureobasidin A (AbA).

Yeast two-hybrid assay (Y2H) was conducted using the Matchmaker Gold Yeast Two-Hybrid System Kit (Takara, Japan, Product ID: 630489). Briefly, the CDS of *CitNAC71* and *CitNAC76* were inserted into pGADT7 and pGBKT7 vectors, respectively. Then, both fusion vectors were co-transformed into Y2HGold competent cells. Single colonies were selected from SD/-Trp/-Leu medium (DDO) and validated on SD/-His/-Leu/-Trp/-Ade medium (QDO) supplemented with AbA and X- α -gal. The sequences of the primers used are listed in [Supplementary Table S8](#).

4.12 Luciferase complementation imaging (LCI) assays

The CDS of *CitNAC71* and *CitNAC76* were cloned into the pCambia1300-nLUC and pCambia1300-cLUC vector, respectively. Both the recombinant plasmid and empty plasmids were introduced into *A. tumefaciens* GV3101 strains, then equal amounts of *A. tumefaciens* cultures carrying nLUC and cLUC constructs were mixed and infiltrated into *N. benthamiana* leaves. After 2 d of incubation, the infiltrated leaves were injected with 0.2 mM D-luciferin (Yeasen, China, Product ID: 40901ES01), and the picture was captured using a chemiluminescent imaging system (Tanon, China). The sequences of the primers used are listed in [Supplementary Table S9](#).

4.13 Bimolecular fluorescence complementation (BiFC) assay

BiFC assay was performed as previously described ([Ni et al. 2023](#)). The CDS of *CitNAC71* and *CitNAC76* were cloned into p2YC and p2YN vectors. *A. tumefaciens* GV3101 harboring the *CitNAC71*-p2YC and *CitNAC76*-p2YN constructs were mixed in equal volumes and infiltrated into *N. benthamiana* leaves. After incubation for 48 h, the YFP signal was detected using a confocal laser scanning microscope (Zeiss, Germany). The excitation/emission wavelengths were set as 514 nm/520-560 nm, respectively. The sequences of the primers used are listed in [Supplementary Table S9](#).

4.14 Pull-down screening coupled with LC-MS/MS

Pull-down screening was performed as described by [Zhang et al. \(2024\)](#). The CitNAC76-GST protein was expressed and bound to the GSTSep Glutathione Agarose Resin (Yeasen, China, Product ID: 20507ES10). After co-incubation with total proteins extracted from citrus callus at 4°C overnight, the protein complexes were eluted using elution buffer containing glutathione. After peptide digestion, the samples were analyzed using liquid chromatography-tandem mass spectrometry (LC-MS/MS).

For the Pull-down assay, the full-length CDS (without stop codon) of *CitNAC71* and *CitNAC76* were cloned into pET32a and pGEX-4T-1 vectors, respectively, to obtain the

fusion proteins of CitNAC71-His and CitNAC76-GST. After co-incubation with CitNAC71-His and bound to the GST-tag agarose resin, the CitNAC76-GST and GST proteins were eluted using elution buffer containing glutathione via gravity column. The protein complexes were further analyzed by immunoblotting using anti-GST and anti-His antibodies (Sangon, China, Product ID: 190101,191001), respectively. Primers used for vector construction are listed in [Supplementary Table S8](#).

4.15 Co-IP assay

The Co-IP assay was conducted by the method of [Li et al. \(2023\)](#). The full-length CDS (without stop codon) of CitNAC71 and CitNAC76 were amplified and cloned into pCambia1301-GFP or pCambia1301-Myc vectors, respectively. The constructs were transformed into *Agrobacterium* strain GV3101 and co-expressed in *N. benthamiana* leaves. After 3 d of incubation, the infiltrated leaves were ground in liquid nitrogen and resuspended in protein extraction buffer (10 mM Tris-HCl [pH = 8.0], 250 mM sucrose, 1% Triton X-100, 10 mM MgCl₂, 5 mM β-mercaptoethanol, 1 mM PMSF and 1% protease inhibitor cocktail). After incubation at 4°C for 2 h, the supernatant was obtained and immunoprecipitated using anti-GFP Nanobody Agarose beads (AlpaLifeBio, China, Product ID: KTSM1301). The protein complexes were washed for five times and then eluted by boiling in SDS-PAGE sample buffer. Finally, the eluted proteins were analyzed by Western blot using anti-Myc and anti-GFP antibodies (Sangon, China, Product ID: D110006, D191040), respectively. The sequences of the primers used are listed in [Supplementary Table S9](#).

4.16 Transient gene expression in citrus leaves and juice sacs

Transient gene expression was carried out in citrus leaves and juice sacs as previously described ([Zhao et al. 2021](#)). For transient overexpression, the CDS of genes (*CitNAC76*, *CitCesA7*, *CitCesA8*, *CitIRX9*, *CitIRX9H* and *CitIRX14H*) were cloned into pBI121 vector. For transient RNAi of *CitNAC76*, a specific sequence (300bp) from CDS of *CitNAC76* was integrated into pGBi vector to form a hairpin structure. Primers are listed in [Supplementary](#)

Table S10. The constructs were transferred into *A. tumefaciens* (strain EHA105) individually, and the *A. tumefaciens* suspensions were prepared using the above method.

For injection of citrus leaves, three-month-old Huyou seedlings were employed for transformation, and opposite leaves situated on the stem were chosen for expression of target genes and empty vector, respectively. For infiltration, the juice sacs of Huyou were collected and immersed in the *A. tumefaciens* suspensions. After infection for 20 min, the juice sacs were rinsed three times with sterile water and cultured on MT medium in a growth chamber (RH 80%, 25°C). Additionally, the juice sacs used for RNAi were subjected to 35°C treatment for 3 d, starting after 2 d of infiltration to induce the expression of *CitNAC76*. Gene transcripts and cell wall components were measured after 5 d of infiltration.

4.17 β -glucuronidase (GUS) staining assays

GUS histochemical staining was performed on transgenic *Arabidopsis* and citrus leaves using a GUS Staining Kit (Coolaber, China, Product ID: SL7160). The prepared plants were collected in GUS staining solution and infiltrated under vacuum, followed by an incubation for 12 h at 37°C. After decolorizing using 70% ethanol 2-3 times, the plants were photographed by a SteREO Discovery.V8 stereomicroscopy (Zeiss, Germany) and a handheld camera.

4.18 Virus-induced gene silencing (VIGS) in Huyou seedlings

TRV-mediated VIGS in citrus was conducted as previously described (Zhao et al. 2021). Briefly, a specific sequence (300 bp) from the CDS of *CitNAC76* was inserted into the TRV2 vector. The TRV1 and TRV2 vectors were introduced into *A. tumefaciens* (strain EHA105). Primers are listed in Supplementary Table S10. *A. tumefaciens* cells were cultured in LB to an OD₆₀₀ of 1.0 and then resuspended in the infiltration buffer described above. The suspension of *A. tumefaciens* harboring TRV1 and TRV2 constructs was mixed 1:1 (v/v) and vacuum-infiltrated into the germinating seeds of Huyou. After rinsing three

times with sterile water, the infected seeds were cultured on the MT medium for 3 d, and then transferred to soil in the growth chamber for about 40 d (Supplementary Fig. S28).

4.19 Transformation and analysis of *Arabidopsis* and tomato

For the genetic transformation of *Arabidopsis*, four-week-old seedlings of *A. thaliana* (Col-0) were infiltrated with the *A. tumefaciens* (GV3101) suspension containing the *CitNAC76*-pBI121 vector via the floral dip method (Clough and Bent 1998). The *A. tumefaciens* (strain GV3101) carrying the *CitNAC76*-pAC004 vector was used for the transformation of tomato (*Solanum lycopersicum* L. cv. MicroTom), as described by Cao et al. (2021). The culture medium for tomato genetic transformation could be found in Supplementary Table S11. T₁-generation and T₂-generation transgenic plants were screened using PCR with Genomic DNA (gDNA) extracted from young leaves (Primers listed in Supplementary Table S12). For each gene, three independent lines (T₂-generation) were used for further determination. RT-qPCR, cell wall measurement, and paraffin sections preparation were as previously described. Five biological replicates with 10 plants/fruit each were used for analysis.

4.20 Stable genetic transformation and analysis in citrus

For citrus transformation, the full-length coding sequence (CDS) of *CitNAC76* was cloned into the pGBi-GFP vector under the control of the CaMV 35S promoter and introduced into *Agrobacterium tumefaciens* (strain EHA105). Stem segments (1-2 cm) derived from sterile seedlings of *Citrus sinensis* (L.) Osbeck cv. Jin Cheng were used as explants and incubated in pre-culture medium. *Agrobacterium* cultures were grown to an OD₆₀₀ of 0.5-0.6, collected by centrifugation, and resuspended in activation medium. Stem segments were pre-cultured and then infected with *Agrobacterium* suspension for 10-15 min. The infected explants were placed on co-cultivation medium and incubated in the dark at 25°C for 3 d. After co-cultivation, explants were transferred to selection medium containing 50 mg L⁻¹ kanamycin and 400 mg L⁻¹ cefotaxime and cultured in the dark at 25°C. After callus formation, explants were subsequently transferred to shoot induction medium

and then to rooting medium. Putative transgenic shoots were screened based on GFP fluorescence and confirmed by PCR using gDNA extracted from young leaves. RT-qPCR, cell wall measurements, and paraffin sectioning were performed as described above. Primers used are listed in [Supplementary Table S10 and S12](#). The composition of all media used is provided in [Supplementary Table S13](#).

4.21 Statistical analysis

Data were statistically analyzed using SPSS (Version 20.0, USA) and represented as mean $\pm$ standard error (SE) in the figures. Statistical significance between two groups was calculated using paired or unpaired two-tailed Student's *t*-test ($*P < 0.05$, $**P < 0.01$ and $***P < 0.001$; ns, no significant differences). Figures were drawn using GraphPad Prism (Version 9.0, USA). The heatmaps of gene expression level were drawn using TBtools ([Chen et al. 2020](#)). Statistical data are provided in Supplementary Data Set S1.

4.22 Accession numbers

Sequence data of all citrus genes mentioned in this study can be found in the Citrus Genome Database (<https://www.citrusgenomedb.org/>) with the following accession numbers: *CitCesA7* (XP_006453581.1), *CitCesA8* (XP_006449537.1), *CitIRX9* (XP_006480661.2), *CitIRX9H* (XP_006450659.1), *CitIRX14H* (XP_006425968.1) *CitNAC76* (XP_006466919.1), *CitNAC71* (XP_006436748.1). Amino acid sequences of CESA, IRX and NAC proteins from different species used in this study were listed in Supplementary Table S3, S4 and S5. Raw RNA-seq reads have been deposited in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (<https://www.ncbi.nlm.nih.gov/sra/>), with the BioProject ID: PRJNA1287125.

Conflicts of interest

The authors declare that they have no conflicts of interest.

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Author contributions

C.D.S conceived and supervised the research. J.P.C designed the research and revised the manuscript. C.K. performed the research, analyzed the data and wrote the manuscript. S.J.Y. conducted the plant transformation, fruit phenotypical analyses and RT-qPCR. C.W.Z performed the pull down-MS, Co-IP and subcellular localization assay. Z.M.Y. analyzed the transcriptome data. X.M.Z. and Y.J.L. collected the phylogenetic data and generated the figure. S.J.L and Y.W. revised the manuscript and acquired funding. Q.L revised the manuscript and extended data. C.N.Z and X.J.L contributed to the development of the approach. D.L.W. collected the experimental sample. All authors read and approved the final manuscript.

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1 Figure legends

2 **Figure 1 Dynamic profiles of cellulose and hemicellulose biosynthesis in citrus** 3 **juice sacs at 15°C and 35°C**

4 (A) Appearance of juice sacs at 15°C and 35°C treatments. Scale bars, 1 cm. (B) The
5 firmness of juice sacs. (C) The whiteness index of juice sacs. (D) Cellulose content
6 in juice sacs. DW, dry weight. (E) Hemicellulose content in juice sacs. (F) *In situ*
7 staining and immunofluorescence labeling of cellulose and hemicellulose in citrus
8 juice sacs. The first and second rows, toluidine blue staining; The third and fourth
9 rows, calcofluor white staining; The fifth and sixth rows, immunofluorescence
10 labeled with LM10 antibody; The seventh and eighth rows, immunofluorescence
11 labeled with LM25 antibody. Scale bars = 200 μ m. (G) Weighted gene co-expression
12 network analysis (WGCNA) of citrus juice sacs at 15°C and 35°C treatments.
13 Module-trait correlations and the corresponding *P*-values are in parentheses. (H)
14 Expression patterns of cellulose and hemicellulose synthesis genes in the
15 candidate module, FPKM values were normalized by row. (I) RT-qPCR analysis of
16 *CitCesA7*, *CitCesA8*, *CitIRX9*, *CitIRX9H* and *CitIRX14H*. The expression levels of 0 d
17 were normalized to 1. *CitACTIN* was amplified as an internal control. Data are
18 presented as mean $\pm$ SE from *n* = 20 (B, C) and *n* = 3 (D, E, and I) biological
19 replicates. Asterisks indicate significant differences compared with 15°C
20 treatments (**P* < 0.05, ***P* < 0.01 and ****P* < 0.001; ns, no significant differences).
21 The *P* values were derived from unpaired two-tailed Student's *t*-test and shown in
22 Supplementary Data Set S1. Open circles in panels (I) represents data points.

23 **Figure 2 CitNAC76 directly activates the expression of *CitCesA7/8* and** 24 ***CitIRX9/9H/14H*.**

25 (A) Regulatory effect of CitNAC76 on promoters of *CitCesA7/8* and *CitIRX9/9H/14H*.
26 Top: schematic diagram of effector and reporter constructs applied in dual-

luciferase reporter assays. Bottom: relative LUC/REN values. The LUC/REN value of the empty vector (SK) was set as 1. (B) The subcellular localization of CitNAC76 in transgenic *Nicotiana benthamiana*, with nucleus marked by mCherry. Scale bar, 50 μm . (C) Interactions between CitNAC76 and the promoters of *CitCesA7/8* and *CitIRX9/9H/14H* by Y1H assay. SD/-Leu, a synthetic defined (SD) medium lacking Leu. AbA150, supplemented with 150 $\text{ng}\cdot\text{mL}^{-1}$ aureobasidin A. (D-I) Electrophoretic mobility shift assay (EMSA) of CitNAC76 protein binding to the promoters of *CitCesA7* (D), *CitCesA8* (E and F), *CitIRX9* (G), *CitIRX9H* (H) and *CitIRX14H* (I). Purified CitNAC76-GST protein was used in EMSA, with empty GST protein as a negative control. The 3' biotin probe was used as binding probe and the non-biotin cold probe as competitor probe. NAC binding sites are underscored, and the mutated bases are indicated in red. The presence or absence of protein/probe is indicated by '+' or '-', respectively. Data are presented as mean $\pm$ SE ($n = 8$). Asterisks indicate significant differences compared with SK group (** $P < 0.01$ and *** $P < 0.001$). The P values were derived from unpaired two-tailed Student's t -test and shown in Supplementary Data Set S1. Open circles in panels (A) represents data points.

Figure 3 Transient overexpression and virus-induced gene silencing (VIGS) of *CitNAC76* in citrus seedlings.

(A) Schematic diagram of transient overexpression in Huyou leaves. Scales bar = 2 cm. (B) GUS staining of leaves after overexpression of *CitNAC76*. Scale bar = 500 μm . (C) The transcript abundance of *CitNAC76* in leaves after 6 d of transient overexpression. (D) Cellulose content in leaves. FW, fresh weight. (E) Hemicellulose content in leaves. (F) Relative expression levels of *CitCesA7/8* and *CitIRX9/9H/14H*. The expression level of the empty vector (EV) was normalized to 1. *CitACTIN* was amplified as an internal control. (G) The growth status of TRV control lines and *CitNAC76*-silencing lines. The citrus seedlings were sampled after growing for 40 d in the greenhouse. Scale bar = 2 cm. (H) Quantifications of *CitNAC76* transcript

levels after VIGS. (I) Cellulose content of seedlings after VIGS. FW, fresh weight. (J) Hemicellulose content of seedlings. (K) *In situ* staining and immunofluorescent labeling of cellulose and hemicellulose in the stem of citrus seedlings. Cellulose was labeled by toluidine blue and calcofluor white, while hemicellulose was labeled by LM10/25 antibody. Bar = 400 μ m. (L) Mean gray values of cell wall fluorescent labeling, including calcofluor white and LM10/25 antibody. Mean gray values were calculated using Image J and represented as AU. (M) Relative expression levels of *CitCesA7/8* and *CitIRX9/9H/14H* after VIGS. The expression levels of TRV control lines were normalized to 1. *CitACTIN* was amplified as an internal control. Data were presented as mean $\pm$ SE from n = 6 (C-F), n = 8 (H-J, M) and n = 12 (L). Asterisks indicate significant differences compared with EV group (C-F) or TRV group (H-M) (* P < 0.05, ** P < 0.01 and *** P < 0.001). The P values were derived from paired two-tailed Student's *t*-test (C-F) and unpaired two-tailed Student's *t*-test (H-M), which are shown in Supplementary Data Set S1. Open circles in panels (C-F, H-J, L and M) represents data points.

Figure 4 Stable overexpression of *CitNAC76* in citrus and tomato.

(A) The phenotype of wild type (WT) and overexpressing *CitNAC76* transgenic citrus lines. *Citrus sinensis* (L.) Osbeck cv. Jin Cheng was adopted for genetic transformation. Scale bar, 2 cm. (B) RT-PCR identification of WT and positive transgenic citrus lines. (C) Quantifications of *CitNAC76* transcript levels in WT and transgenic citrus lines. (D) Cellulose content of WT and transgenic citrus seedlings. The aerial parts of citrus seedlings were used for measurement. FW, fresh weight. (E) Hemicellulose content of citrus seedlings. (F) *In situ* staining and immunofluorescent labeling of cellulose and hemicellulose in the stems of WT and transgenic citrus lines. Cellulose was labeled with toluidine blue and calcofluor white, while hemicellulose was labeled with LM10 and LM25 antibodies. Scale bar, 400 μ m. (G) Relative expression levels of *CitCesA7/8* and *CitIRX9/9H/14H* in WT and transgenic plants. The expression levels of wild type were normalized to 1. *CitACTIN*

was amplified as an internal control. (H) Fruit phenotype of WT and CitNAC76-OE tomato fruits at the red ripe stage. Scale bar, 1 cm. Individual images were placed on a black background for comparison. (I) RT-PCR identification of WT and positive transgenic tomato lines. (J) Quantifications of *CitNAC76* and *SlNAC76* transcript levels in WT and transgenic tomato fruits. N.D., not detected. (K) The firmness of tomato fruits at the red ripe stage. The cross-sections of the tomato fruit were used for measurement. (L) Cellulose content in WT and transgenic tomato fruit. The whole fruit of *Solanum lycopersicum* was used for measurement. DW, dry weight. (M) Hemicellulose content in tomato fruit. (N) *In situ* staining and immunofluorescent labeling of cellulose and hemicellulose in the pericarp of WT and transgenic tomato fruit. Cellulose was labeled by toluidine blue and calcofluor white, while hemicellulose was labeled by LM25 antibody. Scale bar, 100 μ m. (O) Relative expression levels of *SlCesAs* and *SlIRXs* in transgenic and WT plants. The expression levels of wild type were normalized to 1. *SlACTIN* was amplified as an internal control. Data are presented as mean $\pm$ SE from $n=3$ (C-E, G), $n=5$ (J, L, M, and O), and $n=30$ (K). Asterisks indicate significant differences compared with WT ($*P < 0.05$, $**P < 0.01$ and $***P < 0.001$). The P values were derived from unpaired two-tailed Student's *t*-test and shown in Supplementary Data Set S1. Open circles in panels (C-E, G, J-M and O) represents data points.

Figure 5 Functional characterization of CitNAC71 and its regulatory effect on *CitNAC76*

(A) Interaction analysis of CitNAC71 and CitNAC76 by Y2H assay. DDO, synthetic defined (SD) medium lacking Trp and Leu. QDO, SD medium lacking Trp, Leu, His, and Ade. 250 AbA, supplemented with 250 ng·mL⁻¹ aureobasidin A. (B) The interaction between CitNAC71 and CitNAC76 was detected by luciferase complementation imaging (LCI) assays. Pseudo-color scale bar represents the fluorescence intensity. (C) The physical interaction between CitNAC71 and CitNAC76 was confirmed by a bimolecular fluorescence complementation (BiFC)

1 assay. Scales bar, 50 μm . (D) *In vitro* pull-down assay showing the interaction
 2 between CitNAC71 and CitNAC76 protein. CitNAC71-His was incubated with
 3 CitNAC76-GST or GST, and the protein complexes were precipitated by GST-tag
 4 beads and detected by immunoblotting using anti-GST or anti-His antibodies. (E)
 5 Confirmation of the interaction between CitNAC71 and CitNAC76 protein using Co-
 6 IP analysis *in vivo*. GFP, CitNAC76-Myc, and CitNAC71-GFP were transiently
 7 overexpressed in *Nicotiana benthamiana* leaves. GFP-tag beads were used for
 8 immunoprecipitation, and anti-GFP and anti-Myc antibodies were used for
 9 immunoblotting. (F) CitNAC71 attenuates the activation of CitNAC76 on the
 10 *CitCesA7* promoter, which detected by dual-luciferase reporter assays and
 11 luciferase assay. The promoters of *CitCesA7* were used as reporters. The presence
 12 or absence of effectors is indicated by '+' or '-'. The white dashed circle represents
 13 the location where *Agrobacterium* was injected into the *Nicotiana benthamiana*
 14 leaves. Pseudo-color scale bar represents the fluorescence intensity. (G) CitNAC71
 15 attenuates the activation of CitNAC76 on the *CitLRX9* promoter. (H) Analysis of the
 16 regulatory effect of CitNAC71 on CitNAC76 promoter using luciferase assay.
 17 Pseudo-color scale bar represents the fluorescence intensity. (I) Dual-luciferase
 18 reporter assays confirm the suppressive effect of CitNAC71 on the CitNAC76
 19 promoter. The LUC/REN value of the empty vector (SK) was set as 1. (J) Interactions
 20 between CitNAC71 and the CitNAC76 promoter by Y1H assay. SD/-Leu, a synthetic
 21 defined (SD) medium lacking Leu. 100 AbA, supplemented with 100 $\text{ng}\cdot\text{mL}^{-1}$
 22 aureobasidin A. (K-M) Electrophoretic mobility shift assay (EMSA) of CitNAC71
 23 protein with the promoter of CitNAC76. Purified CitNAC71-GST protein was used in
 24 EMSA, with GST protein as a negative control. 3' biotin was used as a binding probe
 25 and cold probe as a competitor. The binding sites are underscored, and the mutated
 26 bases are indicated in red. The presence or absence of protein/probe is indicated by
 27 '+' or '-', respectively. Data are presented as mean $\pm$ SE from six biological replicates
 28 ($n = 6$). Asterisks indicate significant differences based on unpaired two-tailed
 29 Student's *t*-test. (* $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$; ns, no significant

differences). All *P* values are shown in Supplementary Data Set S1. Open circles in panels (F, G and I) represents data points.

Figure 6 CitNAC71 suppresses CitNAC76-induced accumulation of cellulose and hemicellulose in citrus juice sacs.

(A) The appearance of citrus juice sacs after 6 d of overexpression. Scale bar, 1 cm. Citrus juice sacs infiltrated with the empty vector (EV) were used as a negative control. Individual images were placed on a black background for comparison. (B) Quantifications of *CitNAC71* transcript level after overexpression. The expression levels of EV group were normalized to 1. *CitACTIN* was amplified as an internal control. (C) Quantifications of *CitNAC76* transcript level after overexpression. (D) The firmness of juice sacs after transient overexpression. (E) Cellulose content after transient overexpression. DW, dry weight. (F) Hemicellulose content after transient overexpression. (G) *In situ* staining and immunofluorescent labeling of cellulose and hemicellulose in the citrus juice sacs. Cellulose was labeled with toluidine blue and calcofluor white, while hemicellulose was labeled with LM10/25 antibody. Scale bar, 200 μ m. (H) Mean gray values of cell wall fluorescent labeling, including calcofluor white and LM10/25 antibody. Mean gray values were calculated using Image J and are presented as AU. (I) Relative expression levels of *CitCesA7/8* and *CitIRX9/9H/14H* after transient overexpression. The expression levels of EV group were normalized to 1. *CitACTIN* was amplified as an internal control. Data are presented as mean $\pm$ SE from *n* = 3 (B, C, E, F, and I), *n* = 30 (D), and *n* = 12 (H). Asterisks indicate significant differences based on unpaired two-tailed Student's *t*-test. (**P* < 0.05, ***P* < 0.01 and ****P* < 0.001; ns, no significant differences). All *P* values are shown in Supplementary Data Set S1. Open circles in panels (B-F, H and I) represents data points.

Figure 7 Virus-induced gene silencing (VIGS) of CitNAC71 and CitNAC76 in citrus seedlings.

(A) The growth status of control lines and *CitNAC71/CitNAC76* silencing lines. The citrus seedlings were sampled after growing for 40 d in the greenhouse. Scale bar, 2 cm. Citrus seedlings transformed with the TRV empty vector were used as a negative control. (B) Quantifications of *CitNAC71* transcript levels after VIGS. The expression levels of TRV lines were normalized to 1. *CitACTIN* was amplified as an internal control. (C) Quantifications of *CitNAC76* transcript levels after VIGS. (D) The contents of cellulose and hemicellulose after VIGS. FW, fresh weight. (E) *In situ* staining and immunofluorescent labeling of cellulose and hemicellulose in the stem of citrus seedlings. Cellulose was labeled with toluidine blue and calcofluor white, while hemicellulose was labeled with LM10/25 antibody. Scale bar, 400 μ m. (H) Mean gray values of cell wall fluorescent labeling, including calcofluor white and LM10/25 antibody. Mean gray values were calculated using Image J and are presented as AU. (I) Relative expression levels of *CitCesA7/8* and *CitIRX9/9H/14H* after VIGS. The expression levels of TRV control lines were normalized to 1. *CitACTIN* was amplified as an internal control. Data are presented as mean $\pm$ SE from $n = 8$ (B-D, G) and $n = 12$ (F). Asterisks indicate significant differences based on unpaired two-tailed Student's *t*-test. (* $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$; ns, no significant differences). All *P* values are shown in Supplementary Data Set S1. Open circles in panels (B-D, F and G) represents data points.

Figure 8 A proposed regulatory model of the *CitNAC71*-*CitNAC76* module involved in cellulose and hemicellulose biosynthesis in citrus fruit.

At normal temperature, the expression of *CitNAC71* is induced and *CitNAC71* accumulates, causing the inhibition of *CitNAC76* and downregulation of *CitCesA7/8* and *CitIRX9/9H/14H* genes. In contrast, the induction of *CitNAC71* is significantly suppressed under high temperature, allowing *CitNAC76* transcripts to accumulate to high levels, which leads to the upregulation of *CitCesA7/8* and *CitIRX9/9H/14H*. Consequently, excessive accumulation of cellulose and hemicellulose in citrus fruit contributes to the development of granulation disorder. The thickness of the arrows

1 indicates the intensity of the response. The size of the cellulose and hemicellulose
2 molecular structures indicates the amount of cell wall material accumulated. This
3 figure was created with BioRender.com.

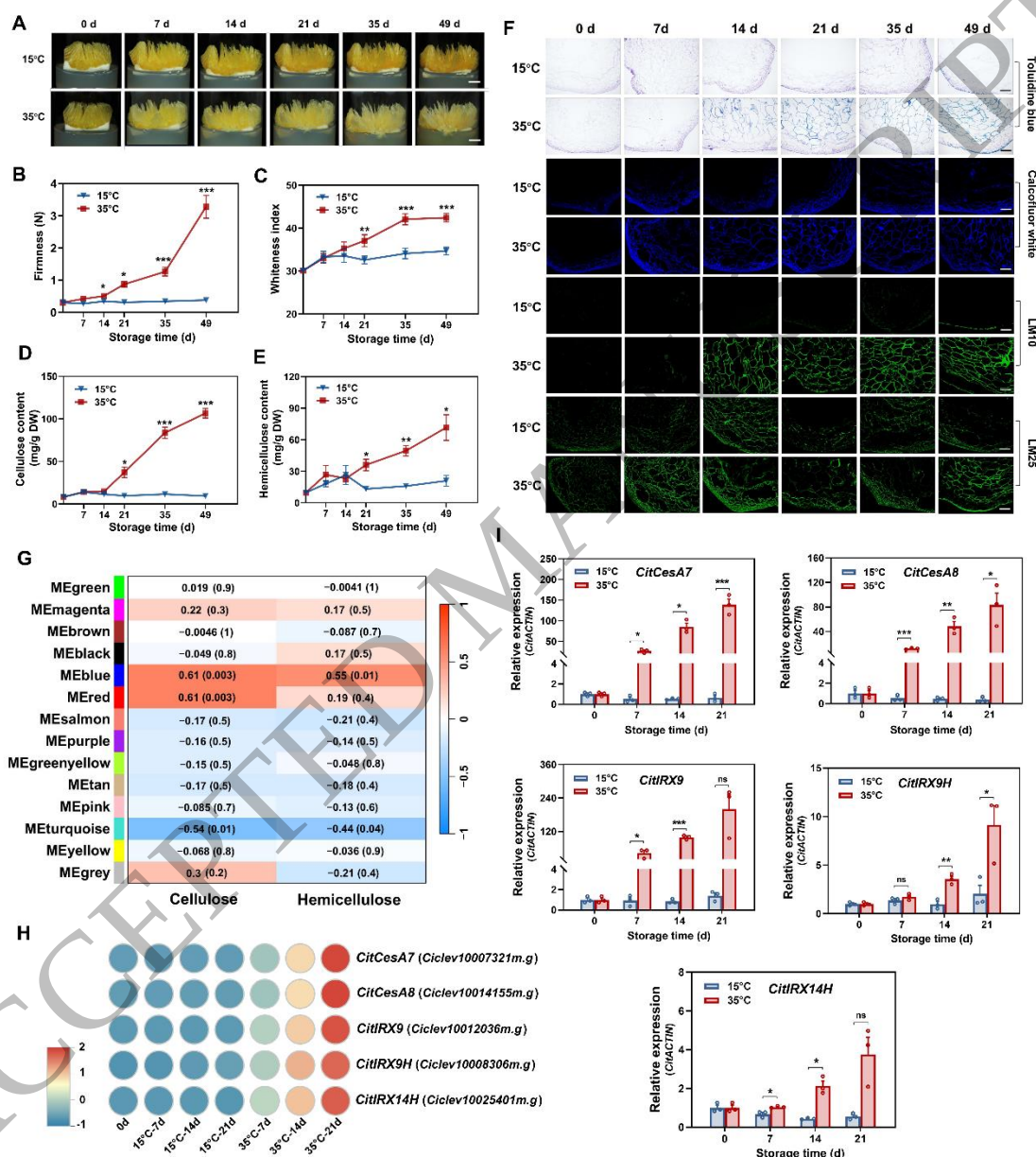


Figure 1
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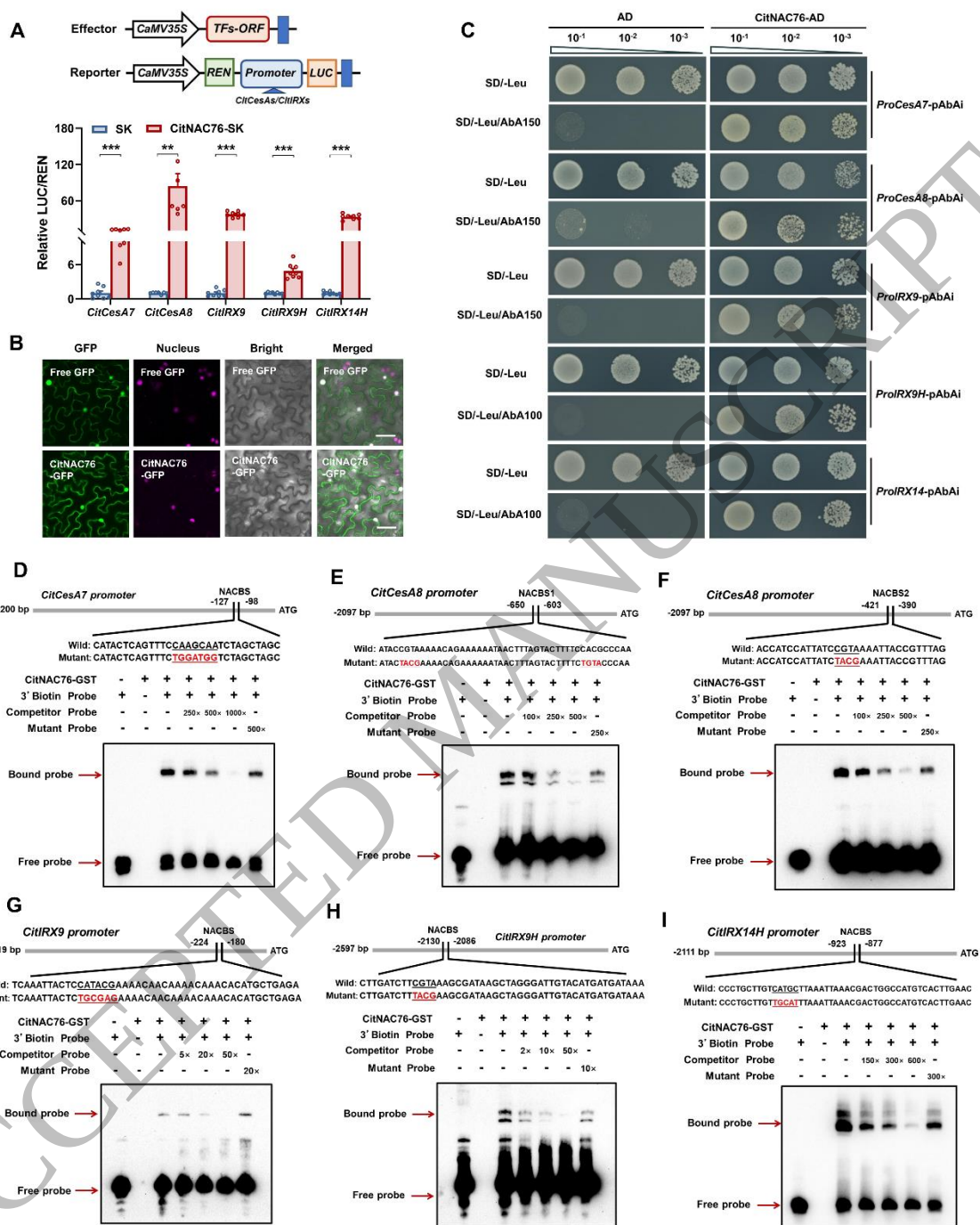


Figure 2
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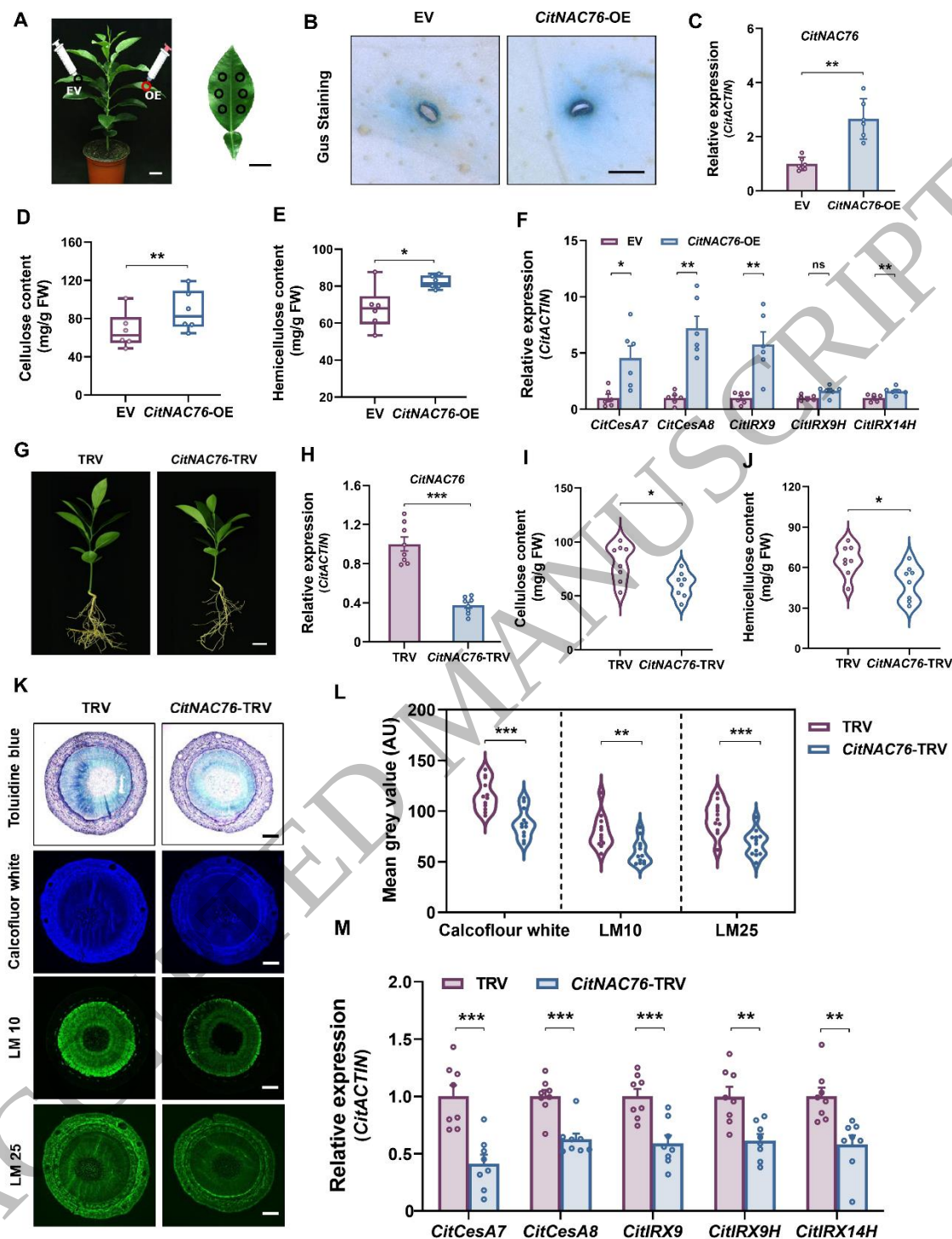


Figure 3

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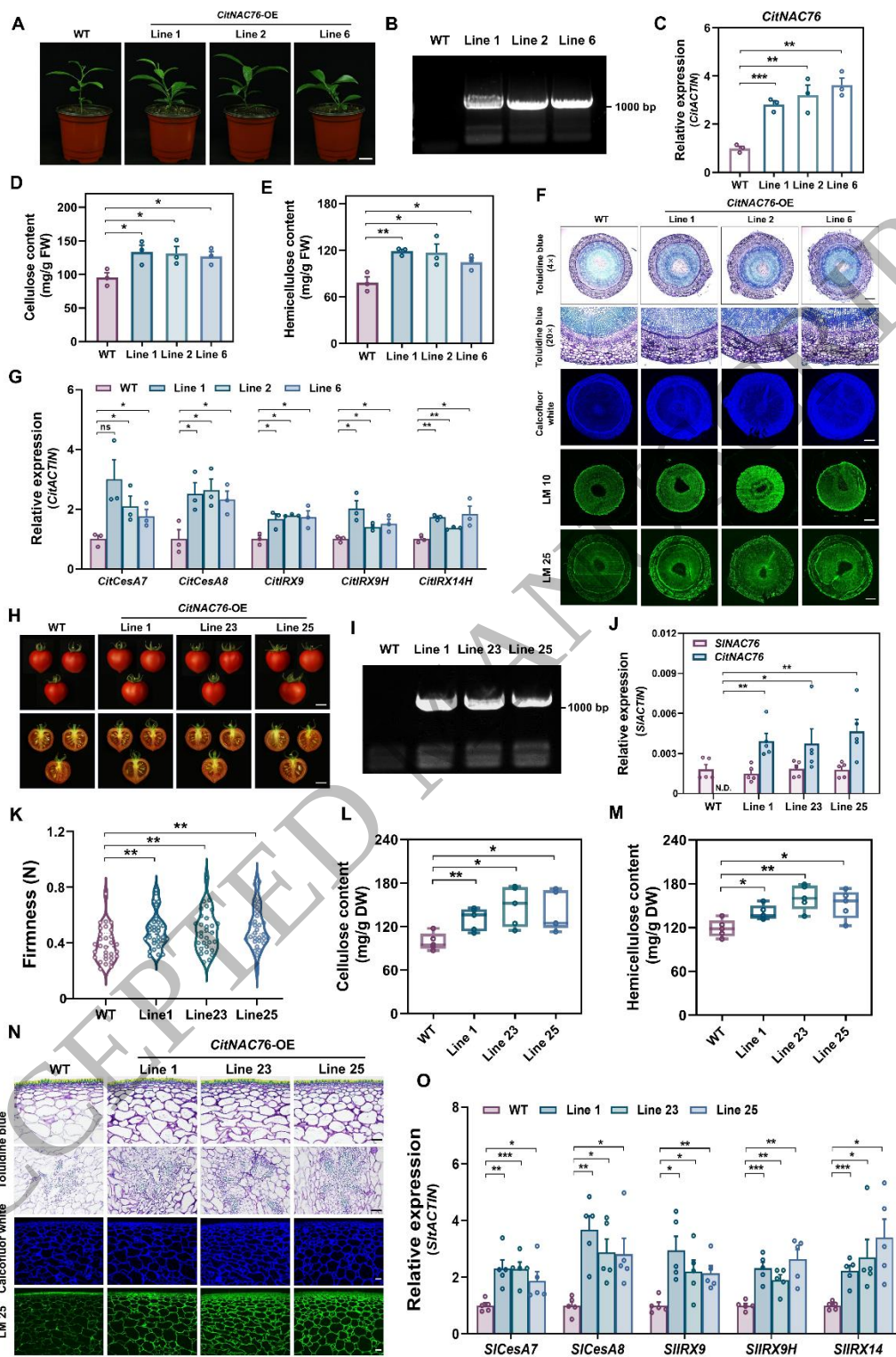


Figure 4
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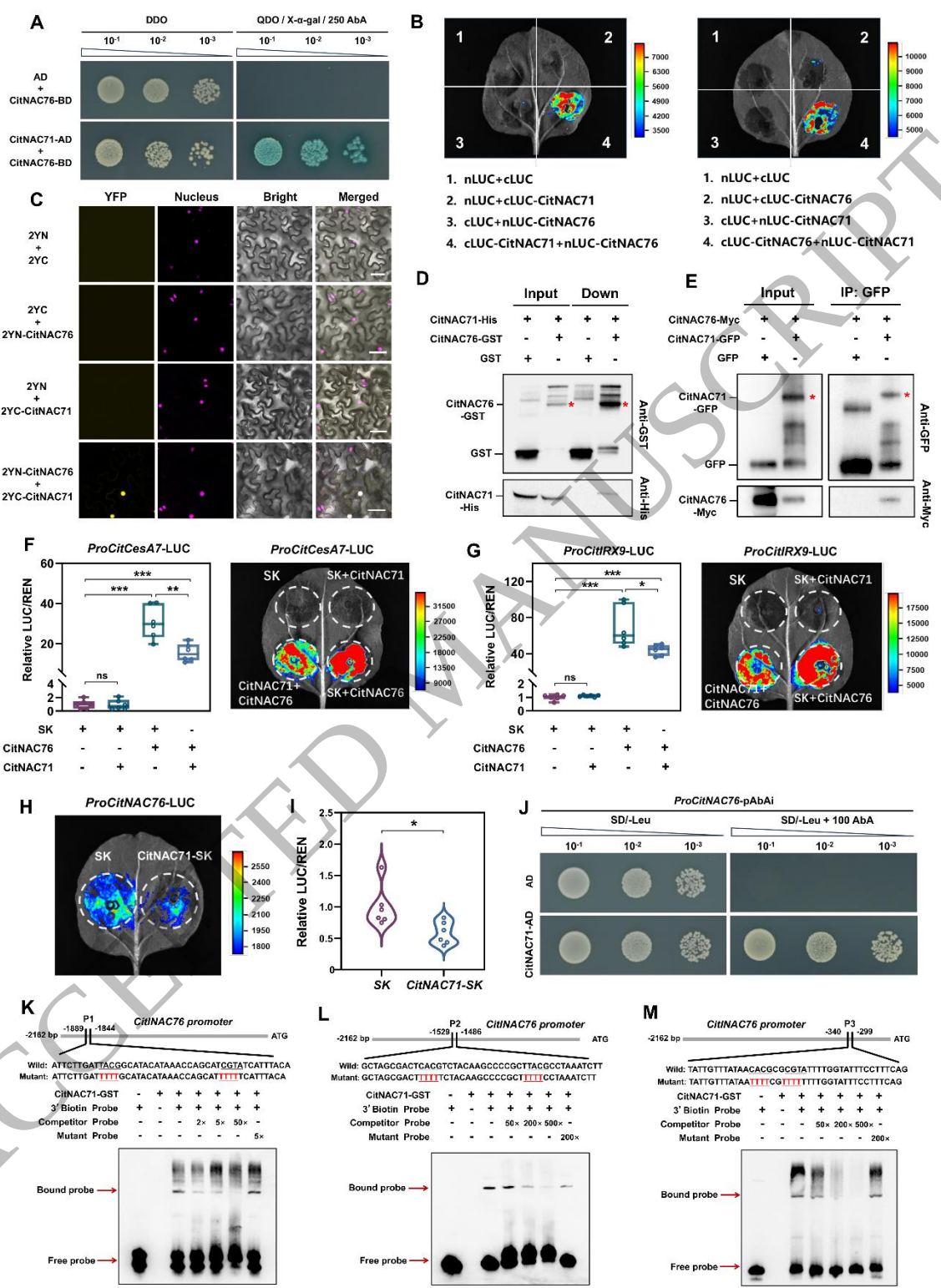


Figure 5
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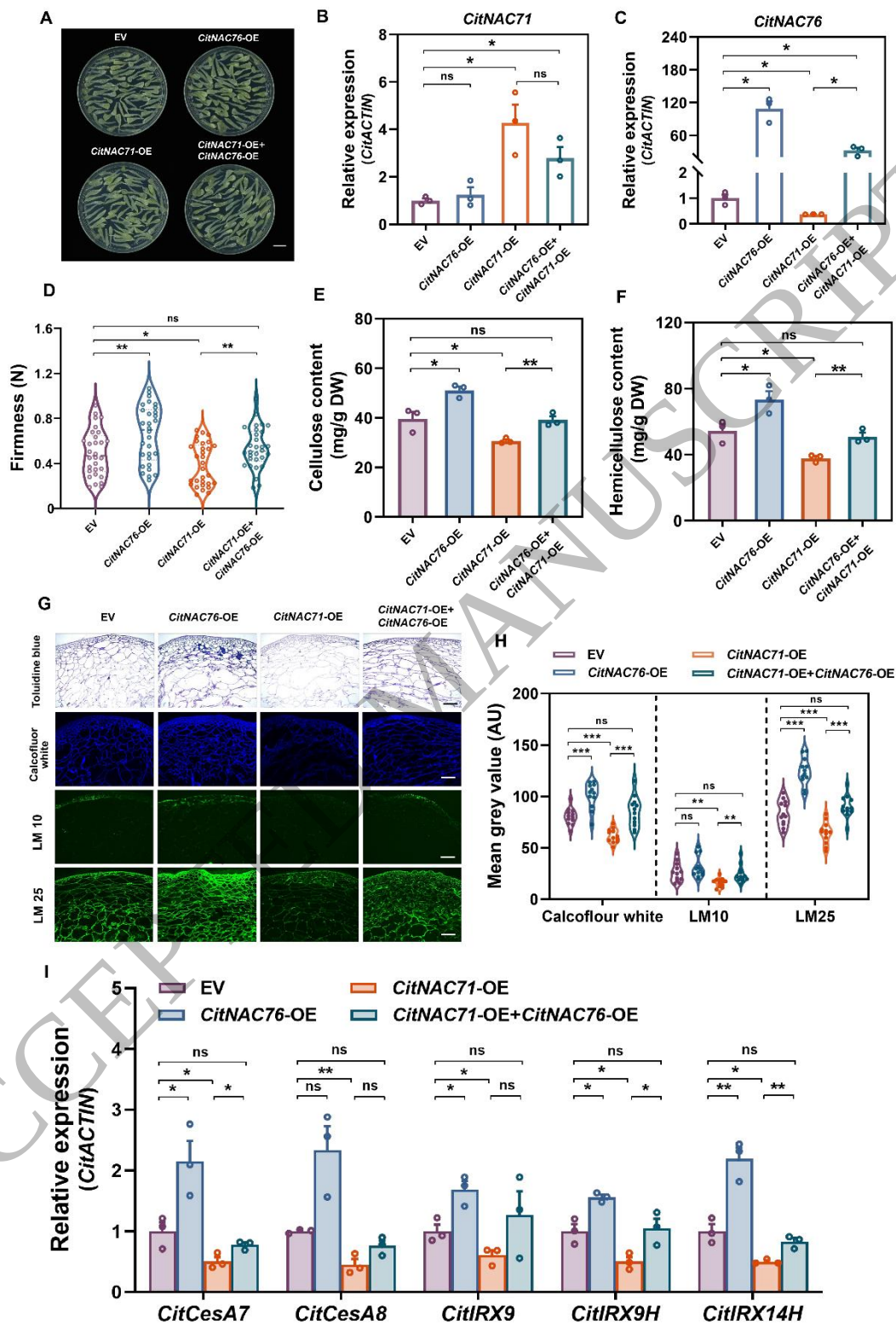


Figure 6

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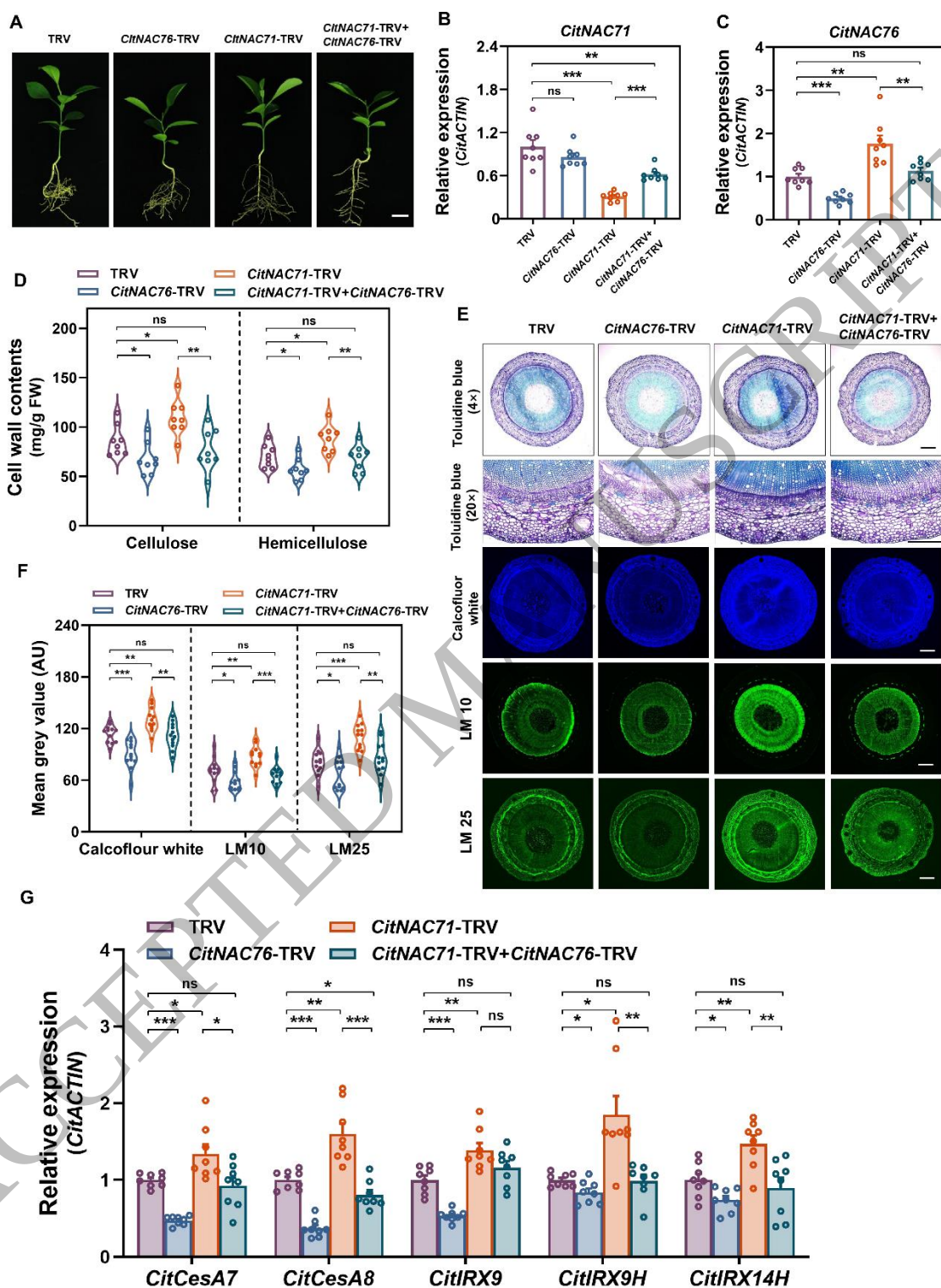


Figure 7
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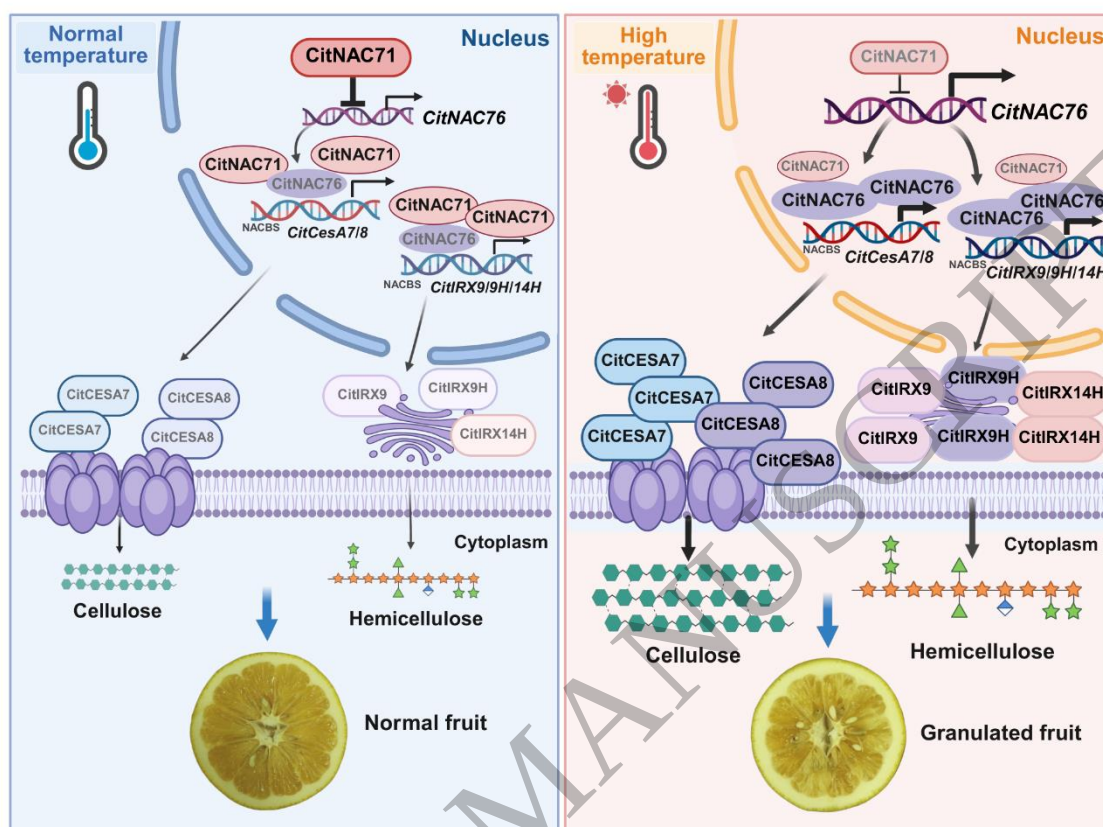


Figure 8
420x559 mm (x DPI)